costs, and is probably performance-driven rather than cost-driven. On this scale the price per kg of a contact lens differs from that of a glass bottle by a factor of 10^5, even though both are made of almost the same glass; the cost per kg of a heart valve differs from that of a plastic bottle by a similar factor, even though both are made of polyethylene. There is obviously something to be learned here.

Look first at the price per unit weight of materials (refer to Figure 17.2). The bulk "commodity" materials of construction and manufacture lie in the shaded band; they all cost between \$0.05 and \$20/kg. Construction materials like concrete, brick, timber, and structural steel lie at the lower end; high-tech materials like titanium alloys lie at the upper. Polymers span a similar range: polyethylene at the bottom, polytetrafluorethylene (PTFE) near the top. Composites lie higher, with GFRP at the bottom and CFRP at the top of the range. Engineering ceramics, at present, lie higher still, though this is changing as production increases. Only the low-volume "exotic" materials lie much above the shaded band.

The price per kg of products (refer to Figure 17.3) shows a different distribution. Eight market sectors are shown, covering much of the manufacturing industry. The shaded band on this figure spans the cost of commodity materials, exactly as on the previous figure. Sectors and their products within the shaded band have the characteristic that material cost is a major fraction of product price: about 50% in civil construction, large marine structures, and some consumer packaging, falling to perhaps 20% as the top of the band is approached (family car—around 25%). The value added in converting material to product in these sectors is relatively low, but the market volume is large. These constraints condition the choice of materials: They must meet modest performance requirements at the lowest possible cost. Associated market sectors generate a driving force for improved processing of conventional materials in order to reduce cost without a loss of performance, or to increase reliability with no increase in cost. For these sectors, incremental improvements in well-tried materials are far more important than revolutionary research findings. Slight improvements in steels, in precision manufacturing methods, or in lubrication technology are quickly assimilated and used.

The products in the upper half of the diagram are technically more sophisticated. The materials of which they are made account for less than 10%—sometimes less than 1%—of the price of the product. The value added to the material during manufacture is high. Product competitiveness is closely linked to material performance. Designers in these sectors have greater freedom in their choice of material and there is a readier acceptance of new materials with attractive property profiles. The market pull here is for

performance, with cost as a secondary consideration. These smaller-volume, higher-value-added sectors drive the development of new or improved materials with enhanced performance: materials that are lighter, or stiffer, or stronger, or tougher, or expand less, or conduct better—or all of these at once.

The sectors have been ordered to form an ascending sequence, prompting the question: What does the horizontal axis measure? Many factors are involved here, one of which can be identified as "information content." The technical knowledge involved in the manufacture of a contact lens or a heart valve is clearly greater than that needed to make a water glass or a plastic bottle. The sectors on the left make few demands on the materials they employ; those on the right push materials to their limits, and at the same time demand the highest reliability. These features make them information-intensive. But there are other factors: market size, competition (or lack of it), perceived value, advertising, fashion, and taste. For this reason, the diagram should not be over-interpreted: It is a help in structuring information, but it is not a quantitative tool.

The manufacturing industry, even in times of recession, has substantial resources; it is in the interests of governments to support their needs. Market pull is ultimately the strongest force for change.

New science: Curiosity-driven research

Curiosity may kill cats, but it is the oxygen of innovative engineering. Technically advanced countries sustain the flow of new ideas by supporting research in three kinds of organization: universities, government laboratories, and industrial research laboratories. Some of the scientists and engineers working in these institutions are given the freedom to pursue ideas that may have no immediate economic objective but may evolve into the materials and manufacturing methods of future decades. Numerous now-commercial materials started in this way. Aluminum, in the time of Napoleon III, was a scientific wonder—he commissioned a set of aluminum spoons for which he paid more than for those of solid silver. Titanium, more recently, has had a similar history. Amorphous (= noncrystalline) metals, now important in transformer technology and in recording heads of tape decks, were, for years, of only academic interest. Semiconductors and superconductors did not appear in response to market forces; it took long-term curiosity-driven research, rewarded in both cases by Nobel prizes, to reveal the principles on which they are based. Polyethylene was discovered by chemists studying the effect of pressure on chemical reactions, not by the sales or marketing departments of multinational corporations. History is dotted with examples of materials and processes that have developed from the inquisitiveness of individuals.

New materials, products of fundamental research, continue to emerge. Some are already entering commercial use, for others the potential is not yet clear. Some, at least, will provide opportunities for innovation; the best may create new markets.

Monolithic ceramics, now produced in commercial quantities, offer high hardness, chemical stability, wear resistance, and resistance to extreme temperatures. Their use as substrates for microcircuits is established, their use in wear-resistant applications is growing, and their use in heat engines is being explored. The emphasis in the development of *composite materials* is shifting toward those that can support loads at higher temperatures. *Metal-matrix composites* (example: aluminum containing particles or fibers of silicon-carbide) and *intermetallic-matrix composites* (titanium-aluminide or molybdenum-disilicide containing silicon-carbide, for instance) can do this. So, potentially, can *ceramic-matrix composites* (alumina with silicon carbide fibers), though the extreme brittleness of these materials requires new design techniques. *Metallic foams*, up to 90% less dense than the parent metal, promise light, stiff sandwich structures competing with composites. *Aerogels*, ultra-low–density foams, provide exceptionally low thermal conductivities.

New *bio-materials*, designed to be implanted in the human body, have structures to which growing tissue will bond without rejection. New *polymers* that can be used at temperatures up to 350°C allow plastics to replace metals in even more applications—the inlet manifold of the automobile engine is an example. New *elastomers* are flexible but strong and tough; they allow better seals, elastic hinges, and resilient coatings. Techniques for producing *functionally graded materials* can give tailored gradients of composition and structure through a component to make it corrosion resistant on the outer surface, tough in the middle, and hard on the inner surface. *Nanostructured materials* promise unique mechanical, electrical, magnetic, and optical properties. *"Intelligent" materials* that can sense and report their condition (via embedded sensors) allow safety margins to be reduced. *Self-healing materials* have the capacity to repair service damage without human intervention.

Developments in *rapid prototyping* now permit single complex parts to be made quickly without dies or molds from a wide range of materials. *Micron-scale fabrication methods* create miniature electro-mechanical systems (MEMS). New techniques of *surface engineering* allow the alloying, coating, or heat-treating of a thin surface layer of a component, modifying its properties to enhance its performance. They include laser hardening, coatings of well-adhering polymers and ceramics, ion implantation, and even the deposition of ultra-hard carbon films with a structure and properties like those of diamond. New *adhesives* displace rivets and spot-welds; the

glue-bonded automobile is a real possibility. And new techniques of *mathematical modeling* and *process control* allow much tighter control of composition and structure in manufacture, reducing cost and increasing reliability and safety.

All these and many more are now realities. They have the potential to enable new designs and to stimulate redesign of products that already have a market, increasing their market share. The designer must stay alert.

17.3 GROWING POPULATION AND WEALTH AND MARKET SATURATION

The world's population continues to grow (see sector 1 in Figure 17.1). Much of this population is also growing in wealth. This wealthier population consumes more products, and as wealth increases they want still more. But thus far the growth of the world's product-producing capacity has grown faster still, with the result that, in developed and developing countries, product markets are saturated. If you want a product—a mobile phone, a refrigerator, a car—you don't have to stand in line with the hope of getting the only one in the store or put your name on a three-year waiting list (as was the case within living memory throughout Europe). Instead, an eager salesperson guides you through the task of selecting, from an array of near-identical products with near-identical prices, the one you think you want.

This has certain consequences. One is the massive and continuing growth in consumption of energy and material resources. This, as a force for change in material selection, was the subject of Chapter 15. Another arises because in a saturated market the designer must seek new ways to attract the consumer. The more traditional reliance on engineering qualities to sell a product are replaced (or augmented) by visual qualities, associations, and carefully managed perceptions created by industrial design. This, too, influences material and process choice; it was the subject of Chapter 16.

17.4 PRODUCT LIABILITY AND SERVICE PROVISION

Legislation now requires that, if a product has a fault, it must be recalled and the fault rectified (see sector 2 in Figure 17.1). Product recalls are exceedingly expensive and they damage the company's image. The higher up the chart shown in Figure 17.3 we go, the more catastrophic a fault becomes. When you buy a packet of six ballpoint pens, it is no great tragedy if one doesn't write properly. The probability ratio 1:6, in a Gaussian distribution, is one standard deviation or "1-sigma." But would you fly in a plane designed

on a "1-sigma" basis? Safety-critical systems today are designed on a "6-sigma" reliability rating, meaning that the probability that the component or assembly will fail to meet specification is less than three parts in a million.

This impacts the way materials are produced and processed. Reliability and reproducibility require sophisticated process control and that every aspect of their production is monitored and verified. Clean steels (steels with greatly reduced inclusion account), aluminum alloys with tight composition control, real-time feedback control of processing, new nondestructive testing for quality and integrity, and random sample testing all help ensure that quality is maintained.

This pressure on manufacturers to assume full-life responsibility for their products causes them to consider taking maintenance and replacement out of the hands of the consumer and doing it themselves. In this way they can monitor product use and replace the product, taking back the original, not when it is worn out but when it is optimal to recondition and redeploy it. The producers no longer sell a product; they sell a service. In common with many others, my university does not own copying machines, though we have many. Instead it has a contract with a provider to guarantee a certain copying service—the provision of "copied pages per week," you might call it.

This changes the economic boundary conditions for the designer of the copier. The objective is no longer that of building copiers with the most features at the lowest price, but rather building ones that will, over the long term, provide copies at the lowest cost per page. The result: a priority to design for ease of replacement, reconditioning, and standardization of materials and components between models. The best copier is no longer the one that lasts the longest; it's the one that is the cheapest for the maker to recondition, upgrade, or replace. This is not an isolated example. Aircraft engine makers now adopt a strategy of providing "power by the hour." It is the same strategy that makers of copiers use: They own the engines, monitor their use, and replace them on the plane at the point at which the tradeoff between cost, reliability, and power-providing capacity is optimal.

Legislation exerts other forces for change. Manufacturers must now register the large number of restricted substances that are contained in the products they make or used in their manufacture. If the level of use exceeds a threshold, the material must be replaced. The list includes the metals cadmium, lead, and mercury, along with a large number of their compounds and chemicals, many of them used in material processing. So the restrictions exert forces for change both in material choice and in the way that the material is shaped, joined, and finished.

17.5 MINIATURIZATION AND MULTIFUNCTIONALITY

Today we make increasing use of precision mechanical engineering at the micron scale (see sector 3 in Figure 17.1). CD and DVD players require the ability to position the read-head with micron precision. Hard disk drives are yet more impressive: Think of storing and retrieving a gigabyte of information on a few square millimeters. MEMS (miniature electro-mechanical systems), now universal as the accelerometers that trigger automobile airbags, rely on micron-thick cantilever beams that bend under inertial forces in a (very) sudden stop. MEMS technology promises much more. There are now video projectors that work not by projecting light through an LCD screen but by reflecting it from arrays of electrostatically actuated mirrors, each a few microns across, allowing much higher light intensities. There are even studies of mechanical microprocessors using bi-stable mechanical switching that could compete with semiconductor switching for information processing.

The smaller you make the components of a device, the greater the functionality you can pack into it. Think (as we already did in Section 8.6) of mobile phones, PDAs, MP3 players, and above all portable computers so small that they fit in a jacket pocket. All these involve components with mechanical functions: protection, positioning, actuation, sensing, casings, hard disks, keyboards, thermal and shock protection. Microengineering no longer means Swiss watches. It means almost all the devices with which we interact continuously throughout a working day.

Miniaturization imposes new demands on materials. As devices become smaller, it is mechanical failures that become design-limiting. In the past the allowable size and weight of casings, connectors, keypads, motors, and actuators gave plenty of margin of safety on bending stiffness, strength, wear rates, and corrosion rates. But none of these scale linearly with size. If we measure scale by a characterizing length L, bending stiffness scales as L^3 or L^4, strength as L^2 or L^3, and wear and corrosion, if measured by fractional loss of section per unit time, scales as $1/L$. Thus the smaller the device, the greater the demands placed on the materials of which it is made.

That is the bad news. The better news is that, since the device is smaller, the amount of material needed to make it is less. Add to that the fact that consumers want small size with powerful functionality and will pay more for it. The result: Expensive, high-performance materials that would not merit consideration for bulky products become viable for this new, miniaturized generation. Think of the armature of an airbag actuator: it weighs about 1 mg, and 1 mg of almost any material—even gold—costs, in it raw state, very little.

So, it seems, material cost no longer limits choice. It is processing that imposes restrictions. Making micron-scale products with precision presents new processing challenges. The watchmakers of Switzerland perfected tools to meet their needs, but the digital watch decimated the mechanical watch business—their methods were not transferable to the new form of miniaturization, which requires mass production at low cost. The makers of microprocessors, on the other hand, had already coped with this sort of scaling-down in nonmechanical systems—their microfabrication methods, rather than those of the watchmaker, have been adapted to make micromachines. But adopting these methods means that the menu of materials is drastically curtailed. In their present form these processes can shape silicon, silicon oxides, nitride and carbide, thin films of copper and gold, but little else. The staple materials of large-scale engineering—carbon and alloy steels, aluminum alloys, polyolefins, glass—do not appear. The current challenge is to broaden the range of microfabrication processes to allow a wider choice of materials to be manipulated.

17.6 CONCERN FOR THE ENVIRONMENT AND FOR THE INDIVIDUAL

We live in a carbon-burning economy. Energy from fossil fuels has given the developed countries high standards of living and made them wealthy. Other, much more populous countries aspire to the same standards of life and wealth and are well-advanced in acquiring them. Yet the time is approaching when burning hydrocarbons will no longer meet energy needs, and even if it could the burden it places on the natural environment will force limits to be set (see sector 4 in Figure 17.1). Short-term measures to address this and the implications these have for material selection were the subject of Chapter 15. The changes necessary to allow longer-term sustainable development will have to be much more far-reaching, changing the ways in which we manufacture, transport ourselves and our goods, and live, and having a major impact on the way materials, central to all of these, are used.

It is one thing to point out that this force for change is a potent one, but quite another to predict what its consequences will be. It is possible to point to fuel-cell technology, energy from natural sources, and safe nuclear power (all with material challenges) as ways forward, but—in the sense we are discussing here—even these are short-term solutions. Remember the figure cited in the introduction to Chapter 15: At a global growth rate of 3% per year, we will mine, process, and dispose of more "stuff" in the next 25 years than in the entire industrial history of humans. The question that has no present answer is this: How can a world population growing at about 3% per year, living in countries with economic growth rates of between 2% and 20% per

year, continue to meet its aspirations with zero—or even negative—growth in consumption of energy and materials?

Wealth has another dimension: It enables resources to be allocated to health care, for which there is a growing demand in a population with an increasing lifespan. A surprisingly large fraction of the organs in a human body perform predominantly mechanical functions: teeth for cutting and grinding, bone to support structural loads, joints to allow articulation, the heart to pump blood, and arteries to carry it under pressure to the extremities of the body, muscle to actuate, skin to provide flexible protection. Aging or accidental damage frequently causes one or more of these to malfunction. Because there are mechanical, it is possible, in principle, to replace them. One way to do so is to use real body parts, but the shear number of patients requiring such treatment and the ethical and other difficulties of using human replacements drives efforts to develop artificial substitutes. Man-made teeth and bone implants, hip and knee joint replacements, artificial artery and skin, even hearts, exist already and are widely used. But at present they are exceedingly expensive, limiting their availability, and for the most part they are only crude substitutes for the real thing. There is major incentive for change here, stimulating research into affordable materials for artificial organs of all types.

Studies show that in a world that is aging the usability of many products and services is unnecessarily challenging to people, whether they are young or old, able-bodied or less able. Until recently the goal of many designers was to appeal to the youth market (meaning the 15–35 age group) as the mainstay of their business, thereby developing products that often could not be used by others. Astute manufacturers have perceived the problem and have sought to redesign their products and services to make them accessible to a wider clientele. The U.S. company OXO has seen sales of its Good Grips range of kitchen and garden tools grow by 50% year after year, and the U.K. supermarket Tesco has added thousands of Internet customers by getting rid of clever but confusing graphics on its website and instead making its services quick to download and the website simple and intuitive to navigate. In this way they include a wider spectrum of people in online shopping. There is a powerful business case for *inclusive design*—design to ensure that products and services address the needs of the widest possible audience. The social case is equally compelling—at present many people are denied use of electronic and other products because they are unable to understand how to use them or lack the motor skills to do so.

All products exclude some users, sometimes deliberately—the childproof medication bottle for instance—but more usually the exclusion is unintended and unnecessary. Many services rely on the use of products for

their delivery, so unusable products deny people access to them too. Video recorders are the standing joke of poor usability, regularly derided in reviews. A much larger number of contemporary products exclude those with disability—limited sight, hearing, physical mobility or strength, or mental acuity. Countering design exclusion is a growing priority of government and is increasingly seen as important by product makers. This thinking influences material choice, changing the constraints and objectives for their selection. The use of materials that, through color or feel, communicate the function of a control, give good grip, insulate, or protect, become priorities.

17.7 SUMMARY AND CONCLUSIONS

Powerful forces drive the development of new and improved materials, encourage substitution, and modify the way in which materials are produced and used. Market forces and military prerogatives, historically the most influential, remain the strongest. The ingenuity of research scientists, too, drives change by revealing a remarkable spectrum of new materials with exciting possibilities, though the time it takes to develop and commercialize them is long: typically 15 years from laboratory to marketplace.

Today additional new drivers influence material development and use. Increasing wealth creates markets for ever more sophisticated products. The trend to products that are smaller in size, lighter in weight, and have greater functionality makes increasing demands on the mechanical properties of the materials used to make them. Greater insistence on product reliability and safety, with the manufacturer held liable for failures or malfunctions, requires materials with consistently reproducible properties and processes that are tightly controlled. Concern for the impact of industrial growth on the natural environment introduces the new objective of selecting materials in such a way as to minimize that impact. And the perception that many products exclude users by their complexity and difficulty of use drives a reassessment of the way in which they are designed and the choice of materials to make them.

The result is that products that were seen as optimal yesterday are no longer optimal today. There is always scope for reassessing designs and the choice of materials to implement them.

17.8 FURTHER READING

Defence and aerospace materials and structures. National Advisory Committee (NAC) Annual Report 2000. *www.iom3.org/foresight/nac/html.*

Ashby, M.F., Ferreira, P.J., & Schodek, D.L. (2009). *Nanomaterials, nanotechnologies and design, an introduction for engineers and architects.* Butterworth-Heinemann, ISBN 978-0-7506-8149-0. *Nanomaterials from an architectural perspective.*

Keates, S., & Clarkson, J. (2004). *Countering design exclusion—An introduction to inclusive design*. Springer-Verlag, ISBN 1-85233-769-9.
An in-depth study of design exclusion and ways to overcome it.

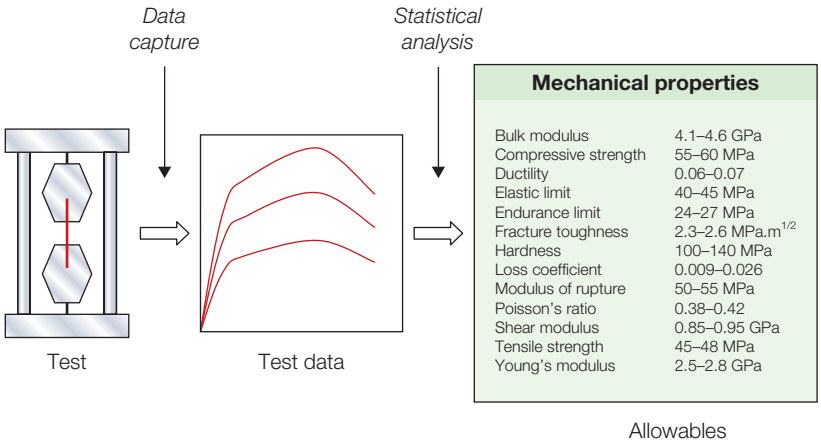
Starke, E.A., & Williams, J.C. (1999). Structural materials: Challenges and opportunities. *The Bridge, 29*(4).
The Bridge *is the journal of the U.S. National Academy of Engineering. This report and the one by Williams can be accessed from their website.*

van Griethuysen, A.J. (Ed.). (1987). *New applications of materials*. Scientific and Technical Publications Ltd, ISBN 0-951362-0-5.
A Dutch study of material evolution.

Williams, J.C. (1998). Engineering competitive materials. *The Bridge, 28*(4).
The Bridge *is the journal of the U.S. National Academy of Engineering. Dr. Williams has long experience in the aerospace industry and in academia.*

This page intentionally left blank

Appendix A: Data for Engineering Materials



Materials Selection in Mechanical Design. DOI: 10.1016/B978-1-85617-663-7.00018-7
© 2011 Michael F. Ashby. Published by Elsevier Ltd. All rights reserved.

CONTENTS

Table A.1 Names and Applications: Metals and Alloys 497
Names and Applications: Polymers and Foams 498
Names and Applications: Composites, Ceramics, Glasses, and Natural Materials 499
Table A.2 Melting Temperature, T_m, and Glass Temperature, T_g . . . 500
Table A.3 Density, ρ 502
Table A.4 Young's Modulus, E 504
Table A.5 Yield Strength, σ_y, and Tensile Strength, σ_{ts} 506
Table A.6 Fracture Toughness (plane strain), K_{IC} 508
Table A.7 Thermal Conductivity, λ, and Thermal Expansion, α . . . 510
Table A.8 Heat Capacity, C_p 512
Table A.9 Resistivity and Dielectric Constant 514
Table A.10 Embodied Energy and CO_2 Footprint 516
Table A.11 Approximate Material Prices, C_m 518
Ways of Checking and Estimating Data 520
Further Reading 523

This appendix lists the names and typical applications common to engineering materials, together with data for their properties.

Table A.1 Names and Applications: Metals and Alloys

Metals	Applications
Ferrous	
Cast irons	Automotive parts, engine blocks, machine tool structural parts, lathe beds
High carbon steels	Cutting tools, springs, bearings, cranks, shafts, railway track
Medium carbon steels	General mechanical engineering (tools, bearings, gears, shafts, bearings)
Low carbon steels	Steel structures ("mild steel")—bridges, oil rigs, ships; reinforcement for concrete; automotive parts, car body panels; galvanized sheet; packaging (cans, drums)
Low alloy steels	Springs, tools, ball bearings, automotive parts (gears connecting rods, etc.)
Stainless steels	Transport, chemical and food processing plant, nuclear plant, domestic ware (cutlery, washing machines, stoves), surgical implements, pipes, pressure vessels, liquid gas containers
Nonferrous	
Aluminum alloys	
Casting alloys	Automotive parts (cylinder blocks), domestic appliances (irons)
Non-heat-treatable alloys	Electrical conductors, heat exchangers, foil, tubes, saucepans, beverage cans, lightweight ships, architectural panels
Heat-treatable alloys	Aerospace engineering, automotive bodies and panels, lightweight structures, ships
Copper alloys	Electrical conductors and wire, electronic circuit boards, heat exchangers, boilers, cookware, coinage, sculptures
Lead alloys	Roof and wall cladding, solder, X-ray shielding, battery electrodes
Magnesium alloys	Automotive castings, wheels, general lightweight castings for transport, nuclear fuel containers; principal alloying addition to aluminum alloys
Nickel alloys	Gas turbines and jet engines, thermocouples, coinage; alloying addition to austenitic stainless steels
Titanium alloys	Aircraft turbine blades; general structural aerospace applications; biomedical implants
Zinc alloys	Die castings (automotive, domestic appliances, toys, handles); coating on galvanized steel

Table A.1 Names and Applications: Polymers and Foams

Polymers	Abbreviation	Applications
Elastomer		
Butyl rubber		Tires, seals, anti-vibration mountings, electrical insulation, tubing
Ethylene-vinyl-acetate	EVA	Bags, films, packaging, gloves, insulation, running shoes
Isoprene	IR	Tires, inner tubes, insulation, tubing, shoes
Natural rubber	NR	Gloves, tires, electrical insulation, tubing
Polychloroprene (neoprene)	CR	Wetsuits, O-rings and seals, footwear
Polyurethane elastomers	el-PU	Packaging, hoses, adhesives, fabric coating
Silicone elastomers		Electrical insulation, electronic encapsulation, medical implants
Thermoplastic		
Acrylonitrile butadiene styrene	ABS	Communication appliances, automotive interiors, luggage, toys, boats
Cellulose polymers	CA	Tool and cutlery handles, decorative trim, pens
Ionomer	I	Packaging, golf balls, blister packs, bottles
Polyamides (nylons)	PA	Gears, bearings, plumbing, packaging, bottles, fabrics, textiles, ropes
Polycarbonate	PC	Safety goggles, shields, helmets, light fittings, medical components
Polyetheretherketone	PEEK	Electrical connectors, racing car parts, fiber composites
Polyethylene	PE	Packaging, bags, squeeze tubes, toys, artificial joints
Polyethylene terephthalate	PET	Blow-molded bottles, film, audio/video tape, sails
Polymethyl methacrylate (acrylic)	PMMA	Aircraft windows, lenses, reflectors, lights, compact discs
Polyoxymethylene (acetal)	POM	Zips, domestic and appliance parts, handles
Polypropylene	PP	Ropes, garden furniture, pipes, kettles, electrical insulation, Astroturf
Polystyrene	PS	Toys, packaging, cutlery, audio cassette/CD cases
Polyurethane thermoplastics	tp-PU	Cushioning, seating, shoe soles, hoses, car bumpers, insulation
Polyvinylchloride	PVC	Pipes, gutters, window frames, packaging
Polytetrafluoroethylene (Teflon)	PTFE	Nonstick coatings, bearings, skis, electrical insulation, tape
Thermoset		
Epoxies	EP	Adhesives, fiber composites, electronic encapsulation
Phenolics	PHEN	Electrical plugs, sockets, cookware, handles, adhesives
Polyester	PEST	Furniture, boats, sporting goods
Polymer foam		
Flexible polymer foam		Packaging, buoyancy, cushioning, sponges, sleeping mats
Rigid polymer foam		Thermal insulation, sandwich panels, packaging, buoyancy

Table A.1 Names and Applications: Composites, Ceramics, Glasses, and Natural Materials

Composites	Applications
Metal polymer	
Aluminum/silicon carbide	Automotive parts, sporting goods
CFRP	Lightweight structural parts (aerospace, bike frames, sporting goods, boat hulls and oars, springs)
GFRP	Boat hulls, automotive parts, chemical plant
Ceramic	
Glass	
Borosilicate glass	Ovenware, laboratory ware, headlights
Glass ceramic	Cookware, lasers, telescope mirrors
Silica glass	High-performance windows, crucibles, high-temperature applications
Soda-lime glass	Windows, bottles, tubing, light bulbs, pottery glazes
Porous	
Brick	Buildings
Concrete	General civil engineering construction
Stone	Buildings, architecture, sculpture
Technical	
Alumina	Cutting tools, spark plugs, microcircuit substrates, valves
Aluminum nitride	Microcircuit substrates and heat sinks
Boron carbide	Lightweight armor, nozzles, dies, precision tool parts
Silicon	Microcircuits, semiconductors, precision instruments, IR windows, MEMS
Silicon carbide	High-temperature equipment, abrasive polishing grits, bearings, armor
Silicon nitride	Bearings, cutting tools, dies, engine parts
Tungsten carbide	Cutting tools, drills, abrasives
Natural	
Bamboo	Building, scaffolding, paper, ropes, baskets, furniture
Cork	Corks and bungs, seals, floats, packaging, flooring
Leather	Shoes, clothing, bags, drive belts
Wood	Construction, flooring, doors, furniture, packaging, sporting goods

Table A.2 Melting Temperature, T_m, and Glass Temperature, T_g

	T_m or T_g (°C)
Metal	
Ferrous	
Cast irons	1130–1250
High carbon steels	1289–1478
Medium carbon steels	1380–1514
Low carbon steels	1480–1526
Low alloy steels	1382–1529
Stainless steels	1375–1450
Nonferrous	
Aluminum alloys	475–677
Copper alloys	982–1082
Lead alloys	322–328
Magnesium alloys	447–649
Nickel alloys	1435–1466
Titanium alloys	1477–1682
Zinc alloys	375–492
Ceramic	
Glass	
Borosilicate glass (*)	450–602
Glass ceramic (*)	563–1647
Silica glass (*)	957–1557
Soda-lime glass (*)	442–592
Porous	
Brick	927–1227
Concrete, typical	927–1227
Stone	1227–1427
Technical	
Alumina	2004–2096
Aluminum nitride	2397–2507
Boron carbide	2372–2507
Silicon	1407–1412
Silicon carbide	2152–2500
Silicon nitride	2388–2496
Tungsten carbide	2827–2920
Composite	
Metal	
Aluminum/silicon carbide	525–627
Polymer	
CFRP	n/a
GFRP	n/a
Natural	
Bamboo (*)	77–102
Cork (*)	77–102

Table A.2 *continued*

	T_m or T_g (°C)
Natural continued	
Leather (*)	107–127
Wood, typical (longitudinal) (*)	77–102
Wood, typical (transverse) (*)	77–102
Polymer	
Elastomer	
Butyl rubber (*)	–73––63
EVA (*)	–73––23
Isoprene (IR) (*)	–83––78
Natural rubber (NR) (*)	–78––63
Neoprene (CR) (*)	–48––43
Polyurethane elastomers (elPU) (*)	–73––23
Silicone elastomers (*)	–123––73
Thermoplastic	
ABS (*)	88–128
Cellulose polymers (CA) (*)	–9–107
Ionomer (I) (*)	27–77
Nylons (PA) (*)	44–56
Polycarbonate (PC) (*)	142–205
PEEK (*)	143–199
Polyethylene (PE) (*)	–25––15
PET (*)	68–80
Acrylic (PMMA) (*)	85–165
Acetal (POM) (*)	–18––8
Polypropylene (PP) (*)	–25––15
Polystyrene (PS) (*)	74–110
Polyurethane thermoplastics (tpPU) (*)	120–160
PVC	75–105
Teflon (PTFE)	107–123
Thermoset	
Epoxies	n/a
Phenolics	n/a
Polyester	n/a
Polymer foam	
Flexible polymer foam (VLD) (*)	112–177
Flexible polymer foam (LD) (*)	112–177
Flexible polymer foam (MD) (*)	112–177
Rigid polymer foam (LD) (*)	67–171
Rigid polymer foam (MD) (*)	67–157
Rigid polymer foam (HD) (*)	67–171

Note: *The table lists the melting point for crystalline solids and the glass temperature for polymeric and inorganic glasses.*

(*) glass transition temperature.

n/a: not applicable (materials decompose rather than melt).

Table A.3 Density, ρ

	ρ (Mg/m^3)
Metal	
Ferrous	
Cast Irons	7.05–7.25
High carbon steels	7.8–7.9
Medium carbon steels	7.8–7.9
Low carbon steels	7.8–7.9
Low alloy steels	7.8–7.9
Stainless steels	7.6–8.1
Nonferrous	
Aluminum alloys	2.5–2.9
Copper alloys	8.93–8.94
Lead alloys	10–11.4
Magnesium alloys	1.74–1.95
Nickel alloys	8.83–8.95
Titanium alloys	4.4–4.8
Zinc alloys	4.95–7
Ceramic	
Glass	
Borosilicate glass	2.2–2.3
Glass ceramic	2.2–2.8
Silica glass	2.17–2.22
Soda-lime glass	2.44–2.49
Porous	
Brick	1.9–2.1
Concrete, typical	2.2–2.6
Stone	2.5–3
Technical	
Alumina	3.5–3.98
Aluminum nitride	3.26–3.33
Boron carbide	2.35–2.55
Silicon	2.3–2.35
Silicon carbide	3–3.21
Silicon nitride	3–3.29
Tungsten carbide	15.3–15.9
Composite	
Metal	
Aluminum/silicon carbide	2.66–2.9
Polymer	
CFRP	1.5–1.6
GFRP	1.75–1.97

Table A.3 *continued*

	ρ (Mg/m^3)
Natural	
Bamboo	0.6–0.8
Cork	0.12–0.24
Leather	0.81–1.05
Wood, typical (longitudinal)	0.6–0.8
Wood, typical (transverse)	0.6–0.8
Polymer	
Elastomer	
Butyl rubber	0.9–0.92
EVA	0.945–0.955
Isoprene (IR)	0.93–0.94
Natural rubber (NR)	0.92–0.93
Neoprene (CR)	1.23–1.25
Polyurethane elastomers (elPU)	1.02–1.25
Silicone elastomers	1.3–1.8
Thermoplastic	
ABS	1.01–1.21
Cellulose polymers (CA)	0.98–1.3
Ionomer (I)	0.93–0.96
Nylons (PA)	1.12–1.14
Polycarbonate (PC)	1.14–1.21
PEEK	1.3–1.32
Polyethylene (PE)	0.939–0.96
PET	1.29–1.4
Acrylic (PMMA)	1.16–1.22
Acetal (POM)	1.39–1.43
Polypropylene (PP)	0.89–0.91
Polystyrene (PS)	1.04–1.05
Polyurethane thermoplastics (tpPU)	1.12–1.24
PVC	1.3–1.58
Teflon (PTFE)	2.14–2.2
Thermoset	
Epoxies	1.11–1.4
Phenolics	1.24–1.32
Polyester	1.04–1.4
Polymer foam	
Flexible polymer foam (VLD)	0.016–0.035
Flexible polymer foam (LD)	0.038–0.07
Flexible polymer foam (MD)	0.07–0.115
Rigid polymer foam (LD)	0.036–0.07
Rigid polymer foam (MD)	0.078–0.165
Rigid polymer foam (HD)	0.17–0.47

Table A.4 Young's Modulus, E

	E (GPa)
Metal	
Ferrous	
Cast irons	165–180
High carbon steels	200–215
Medium carbon steels	200–216
Low carbon steels	200–215
Low alloy steels	201–217
Stainless steels	189–210
Nonferrous	
Aluminum alloys	68–82
Copper alloys	112–148
Lead alloys	12.5–15
Magnesium alloys	42–47
Nickel alloys	190–220
Titanium alloys	90–120
Zinc alloys	68–95
Ceramic	
Glass	
Borosilicate glass	61–64
Glass ceramic	64–110
Silica glass	68–74
Soda-lime glass	68–72
Porous	
Brick	15–25
Concrete, typical	25–38
Stone	20–60
Technical	
Alumina	215–413
Aluminum nitride	302–348
Boron carbide	400–472
Silicon	140–155
Silicon carbide	300–460
Silicon nitride	280–310
Tungsten carbide	600–720
Composite	
Metal	
Aluminum/silicon carbide	81–100
Polymer	
CFRP	69–150
GFRP	15–28

Table A.4 *continued*

	E (GPa)
Natural	
Bamboo	15–20
Cork	0.013–0.05
Leather	0.1–0.5
Wood, typical (longitudinal)	6–20
Wood, typical (transverse)	0.5–3
Polymer	
Elastomer	
Butyl rubber	0.001–0.002
EVA	0.01–0.04
Isoprene (IR)	0.0014–0.004
Natural rubber (NR)	0.0015–0.0025
Neoprene (CR)	0.0007–0.002
Polyurethane elastomers (elPU)	0.002–0.003
Silicone elastomers	0.005–0.02
Thermoplastic	
ABS	1.1–2.9
Cellulose polymers (CA)	1.6–2
Ionomer (I)	0.2–0.424
Nylons (PA)	2.62–3.2
Polycarbonate (PC)	2–2.44
PEEK	3.5–4.2
Polyethylene (PE)	0.621–0.896
PET	2.76–4.14
Acrylic (PMMA)	2.24–3.8
Acetal (POM)	2.5–5
Polypropylene (PP)	0.896–1.55
Polystyrene (PS)	2.28–3.34
Polyurethane thermoplastics (tpPU)	1.31–2.07
PVC	2.14–4.14
Teflon (PTFE)	0.4–0.552
Thermoset	
Epoxies	2.35–3.075
Phenolics	2.76–4.83
Polyester	2.07–4.41
Polymer foam	
Flexible polymer foam (VLD)	0.0003–0.001
Flexible polymer foam (LD)	0.001–0.003
Flexible polymer foam (MD)	0.004–0.012
Rigid polymer foam (LD)	0.023–0.08
Rigid polymer foam (MD)	0.08–0.2
Rigid polymer foam (HD)	0.2–0.48

Table A.5 Yield Strength, σ_y, and Tensile Strength, σ_{ts}

	σ_y (MPa)	σ_{ts} (MPa)
Metal		
Ferrous		
Cast irons	215–790	350–1000
High carbon steels	400–1155	550–1640
Medium carbon steels	305–900	410–1200
Low carbon steels	250–395	345–580
Low alloy steels	400–1100	460–1200
Stainless steels	170–1000	480–2240
Nonferrous		
Aluminum alloys	30–500	58–550
Copper alloys	30–500	100–550
Lead alloys	8–14	12–20
Magnesium alloys	70–400	185–475
Nickel alloys	70–1100	345–1200
Titanium alloys	250–1245	300–1625
Zinc alloys	80–450	135–520
Ceramic		
Glass		
Borosilicate glass (*)	264–384	22–32
Glass ceramic (*)	750–2129	62–177
Silica glass (*)	1100–1600	45–155
Soda-lime glass (*)	360–420	31–35
Porous		
Brick (*)	50–140	7–14
Concrete, typical (*)	32–60	2–6
Stone (*)	34–248	5–17
Technical		
Alumina (*)	690–5500	350–665
Aluminum nitride (*)	1970–2700	197–270
Boron carbide (*)	2583–5687	350–560
Silicon (*)	3200–3460	160–180
Silicon carbide (*)	1000–5250	370–680
Silicon nitride (*)	524–5500	690–800
Tungsten carbide (*)	3347–6833	370–550
Composite		
Metal		
Aluminum/silicon carbide	280–324	290–365
Polymer		
CFRP	550–1050	550–1050
GFRP	110–192	138–241

Table A.5 *continued*

	σ_y (MPa)	σ_{ts} (MPa)
Natural		
Bamboo	35–44	36–45
Cork	0.3–1.5	0.5–2.5
Leather	5–10	20–26
Wood, typical (longitudinal)	30–70	60–100
Wood, typical (transverse)	2–6	4–9
Polymer		
Elastomer		
Butyl rubber	2–3	5–10
EVA	12–18	16–20
Isoprene (IR)	20–25	20–25
Natural rubber (NR)	20–30	22–32
Neoprene (CR)	3.4–24	3.4–24
Polyurethane elastomers (elPU)	25–51	25–51
Silicone elastomers	2.4–5.5	2.4–5.5
Thermoplastic		
ABS	18.5–51	27.6–55.2
Cellulose polymers (CA)	25–45	25–50
Ionomer (I)	8.3–15.9	17.2–37.2
Nylons (PA)	50–94.8	90–165
Polycarbonate (PC)	59–70	60–72.4
PEEK	65–95	70–103
Polyethylene (PE)	17.9–29	20.7–44.8
PET	56.5–62.3	48.3–72.4
Acrylic (PMMA)	53.8–72.4	48.3–79.6
Acetal (POM)	48.6–72.4	60–89.6
Polypropylene (PP)	20.7–37.2	27.6–41.4
Polystyrene (PS)	28.7–56.2	35.9–56.5
Polyurethane thermoplastics (tpPU)	40–53.8	31–62
PVC	35.4–52.1	40.7–65.1
Teflon (PTFE)	15–25	20–30
Thermoset		
Epoxies	36–71.7	45–89.6
Phenolics	27.6–49.7	34.5–62.1
Polyester	33–40	41.4–89.6
Polymer foam		
Flexible polymer foam (VLD)	0.01–0.12	0.24–0.85
Flexible polymer foam (LD)	0.02–0.3	0.24–2.35
Flexible polymer foam (MD)	0.05–0.7	0.43–2.95
Rigid polymer foam (LD)	0.3–1.7	0.45–2.25
Rigid polymer foam (MD)	0.4–3.5	0.65–5.1
Rigid polymer foam (HD)	0.8–12	1.2–12.4

(*) NB: For ceramics, yield strength is replaced by compressive strength, *which is more relevant in ceramic design. Note that ceramics are of the order of 10 times stronger in compression than in tension.*

Table A.6 Fracture Toughness (plane strain), K_{IC}

	K_{IC} (MPa$\sqrt{\text{m}}$)
Metal	
Ferrous	
Cast irons	22–54
High carbon steels	27–92
Medium carbon steels	12–92
Low carbon steels	41–82
Low alloy steels	14–200
Stainless steels	62–280
Nonferrous	
Aluminum alloys	22–35
Copper alloys	30–90
Lead alloys	5–15
Magnesium alloys	12–18
Nickel alloys	80–110
Titanium alloys	14–120
Zinc alloys	10–100
Ceramic	
Glass	
Borosilicate glass	0.5–0.7
Glass ceramic	1.4–1.7
Silica glass	0.6–0.8
Soda-lime glass	0.55–0.7
Porous	
Brick	1–2
Concrete, typical	0.35–0.45
Stone	0.7–1.5
Technical	
Alumina	3.3–4.8
Aluminum nitride	2.5–3.4
Boron carbide	2.5–3.5
Silicon	0.83–0.94
Silicon carbide	2.5–5
Silicon nitride	4–6
Tungsten carbide	2–3.8
Composite	
Metal	
Aluminum/silicon carbide	15–24
Polymer	
CFRP	6.1–88
GFRP	7–23

Table A.6 *continued*

	K_{IC} (MPa$\sqrt{m}$)
Natural	
Bamboo	5–7
Cork	0.05–0.1
Leather	3–5
Wood, typical (longitudinal)	5–9
Wood, typical (transverse)	0.5–0.8
Polymer	
Elastomer	
Butyl rubber	0.07–0.1
EVA	0.5–0.7
Isoprene (IR)	0.07–0.1
Natural rubber (NR)	0.15–0.25
Neoprene (CR)	0.1–0.3
Polyurethane elastomers (elPU)	0.2–0.4
Silicone elastomers	0.03–0.5
Thermoplastic	
ABS	1.19–4.30
Cellulose polymers (CA)	1–2.5
Ionomer (I)	1.14–3.43
Nylons (PA)	2.22–5.62
Polycarbonate (PC)	2.1–4.60
PEEK	2.73–4.30
Polyethylene (PE)	1.44–1.72
PET	4.5–5.5
Acrylic (PMMA)	0.7–1.6
Acetal (POM)	1.71–4.2
Polypropylene (PP)	3–4.5
Polystyrene (PS)	0.7–1.1
Polyurethane thermoplastics (tpPU)	1.84–4.97
PVC	1.46–5.12
Teflon (PTFE)	1.32–1.8
Thermoset	
Epoxies	0.4–2.22
Phenolics	0.79–1.21
Polyester	1.09–1.70
Polymer foam	
Flexible polymer foam (VLD)	0.005–0.02
Flexible polymer foam (LD)	0.015–0.05
Flexible polymer foam (MD)	0.03–0.09
Rigid polymer foam (LD)	0.002–0.02
Rigid polymer foam (MD)	0.007–0.049
Rigid polymer foam (HD)	0.024–0.091

Note: K_{IC} is only valid for conditions under which linear elastic fracture mechanics apply (see Chapter 8). The plane-strain toughness, G_{IC}, may be estimated from $K_{IC}^2 = E G_{IC} / (1 - \nu^2) \approx E G_{IC}$ (as $\nu^2 \approx 0.1$).

Table A.7 Thermal Conductivity, λ, and Thermal Expansion, α

	λ (W/m.K)	α (10^{-6}/C)
Metal		
Ferrous		
Cast irons	29–44	10–12.5
High carbon steels	47–53	11–13.5
Medium carbon steels	45–55	10–14
Low carbon steels	49–54	11.5–13
Low alloy steels	34–55	10.5–13.5
Stainless steels	11–19	13–20
Nonferrous		
Aluminum alloys	76–235	21–24
Copper alloys	160–390	16.9–18
Lead alloys	22–36	18–32
Magnesium alloys	50–156	24.6–28
Nickel alloys	67–91	12–13.5
Titanium alloys	5–12	7.9–11
Tungsten alloys	100–142	4–5.6
Zinc alloys	100–135	23–28
Ceramic		
Glass		
Borosilicate glass	1–1.3	3.2–4.0
Glass ceramic	1.3–2.5	1–5
Silica glass	1.4–1.5	0.55–0.75
Soda-lime glass	0.7–1.3	9.1–9.5
Porous		
Brick	0.46–0.73	5–8
Concrete, typical	0.8–2.4	6–13
Stone	5.4–6.0	3.7–6.3
Technical		
Alumina	30–38.5	7–10.9
Aluminum nitride	80–200	4.9–6.2
Boron carbide	40–90	3.2–3.4
Silicon	140–150	2.2–2.7
Silicon carbide	115–200	4.0–5.1
Silicon nitride	22–30	3.2–3.6
Tungsten carbide	55–88	5.2–7.1
Composite		
Metal		
Aluminum/silicon carbide	180–160	15–23
Polymer		
CFRP	1.28–2.6	1–4
GFRP	0.4–0.55	8.6–33

Table A.7 *continued*

	λ (W/m.K)	α (10^{-6}/C)
Natural		
Bamboo	0.1–0.18	2.6–10
Cork	0.035–0.048	130–230
Leather	0.15–0.17	40–50
Wood, typical (longitudinal)	0.31–0.38	2–11
Wood, typical (transverse)	0.15–0.19	32–42
Polymer		
Elastomer		
Butyl rubber	0.08–0.1	120–300
EVA	0.3–0.4	160–190
Isoprene (IR)	0.08–0.14	150–450
Natural rubber (NR)	0.1–0.14	150–450
Neoprene (CR)	0.08–0.14	575–610
Polyurethane elastomers	0.28–0.3	150–165
Silicone elastomers	0.3–1.0	250–300
Thermoplastic		
ABS	0.19–0.34	84.6–234
Cellulose polymers (ca)	0.13–0.3	150–300
Ionomer (I)	0.24–0.28	180–306
Nylons (PA)	0.23–0.25	144–150
Polycarbonate (PC)	0.19–0.22	120–137
PEEK	0.24–0.26	72–194
Polyethylene (PE)	0.40–0.44	126–198
PET	0.14–0.15	114–120
Acrylic (PMMA)	0.08–0.25	72–162
Acetal (POM)	0.22–0.35	76–201
Polypropylene (PP)	0.11–0.17	122–180
Polystyrene (PS)	0.12–0.13	90–153
Polyurethane thermoplastics	0.23–0.24	90–144
PVC	0.15–0.29	100–150
Teflon (PTFE)	0.24–0.26	126–216
Thermoset		
Epoxies	0.18–0.5	58–117
Phenolics	0.14–0.15	120–125
Polyester	0.28–0.3	99–180
Polymer foam		
Flexible polymer foam (VLD)	0.036–0.048	120–220
Flexible polymer foam (LD)	0.04–0.06	115–220
Flexible polymer foam (MD)	0.04–0.08	115–220
Rigid polymer foam (LD)	0.023–0.04	20–80
Rigid polymer foam (MD)	0.027–0.038	20–75
Rigid polymer foam (HD)	0.034–0.06	22–70

Table A.8 Heat Capacity, C_p

	C_p (J/kg.K)
Metal	
Ferrous	
Cast irons	439–495
High carbon steels	440–510
Medium carbon steels	440–510
Low carbon steels	460–505
Low alloy steels	410–530
Stainless steels	450–530
Nonferrous	
Aluminum alloys	857–990
Copper alloys	372–388
Lead alloys	122–145
Magnesium alloys	955–1060
Nickel alloys	452–460
Titanium alloys	520–600
Zinc alloys	405–535
Ceramic	
Glass	
Borosilicate glass	760–800
Glass ceramic	600–900
Silica glass	680–730
Soda-lime glass	850–950
Porous	
Brick	750–850
Concrete, typical	835–1050
Stone	840–920
Technical	
Alumina	790–820
Aluminum nitride	780–820
Boron carbide	840–1290
Silicon	668–715
Silicon carbide	663–800
Silicon nitride	670–800
Tungsten carbide	184–292
Composite	
Metal	
Aluminum/silicon carbide	800–900
Polymer	
CFRP	900–1040
GFRP	1000–1200

Table A.8 *continued*

	C_p (J/kg.K)
Natural	
Bamboo	1660–1710
Cork	1900–2100
Leather	1530–1730
Wood, typical	1660–1710
Aluminum nitride	780–820
Polymer	
Elastomer	
Butyl rubber	1800–2500
EVA	2000–2200
Isoprene (IR)	1800–2500
Natural rubber (NR)	1800–2500
Neoprene (CR)	2000–2200
Polyurethane elastomers (elPU)	1650–1700
Silicone elastomers	1050–1300
Thermoplastic	
ABS	1390–1920
Cellulose polymers (CA)	1390–1670
Ionomer (I)	1810–1890
Nylons (PA)	1600–1660
Polycarbonate (PC)	1530–1630
PEEK	1440–1500
Polyethylene (PE)	1810–1880
PET	1420–1470
Acrylic (PMMA)	1490–1610
Acetal (POM)	1360–1430
Polypropylene (PP)	1870–1960
Polystyrene (PS)	1690–1760
Polyurethane thermoplastics (tpPU)	1550–1620
PVC	1360–1440
Teflon (PTFE)	1010–1050
Thermoset	
Epoxies	1490–2000
Phenolics	1470–1530
Polyester	1510–1570
Polymer foam	
Flexible polymer foam (VLD)	1750–2260
Flexible polymer foam (LD)	1750–2260
Flexible polymer foam (MD)	1750–2260
Rigid polymer foam (LD)	1120–1910
Rigid polymer foam (MD)	1120–1910
Rigid polymer foam (HD)	1120–1910

Table A.9 Resistivity and Dielectric Constant

	Resistivity (μohm.cm)	Dielectric Constant
Metal		
Ferrous		
Cast irons	49–56	–
High carbon steels	17–20	–
Medium carbon steels	15–22	–
Low carbon steels	15–20	–
Low alloy steels	15–35	–
Stainless steels	64–107	–
Nonferrous		
Aluminum alloys	2.5–5.0	–
Copper alloys	1.7–24	–
Lead alloys	15–22	–
Magnesium alloys	4.2–15	–
Nickel alloys	6–114	–
Titanium alloys	100–170	–
Tungsten alloys	10.2–14	–
Zinc alloys	5.4–7.2	–
Ceramic		
Glass		
Borosilicate glass	3×10^{21}–3×10^{22}	4.6–6.0
Glass ceramic	2×10^{19}–1×10^{21}	5.3–6.2
Silica glass	1×10^{23}–1×10^{27}	3.7–3.9
Soda-lime glass	8×10^{17}–8×10^{18}	7.0–7.6
Porous		
Brick	1×10^{14}–3×10^{16}	7.0–10
Concrete, typical	1.8×10^{12}–1.8×10^{13}	8.0–12
Stone	1×10^{8}–1×10^{14}	6.0–18
Technical		
Alumina	1×10^{20}–1×10^{22}	6.5–6.8
Aluminum nitride	1×10^{19}–1×10^{21}	8.3–9.3
Boron carbide	1×10^{5}–1×10^{7}	4.8–8.0
Silicon	1×10^{6}–1×10^{12}	11–12
Silicon carbide	1×10^{9}–1×10^{12}	6.3–9.0
Silicon nitride	1×10^{20}–1×10^{21}	7.9–8.1
Tungsten carbide	20–100	–
Composite		
Metal		
Aluminum/silicon carbide	5–12	–
Polymer		
CFRP	1.7×10^{5}–1×10^{6}	–
GFRP	1×10^{16}–2×10^{22}	4.2–5.2

Table A.9 *continued*

	Resistivity (μohm.cm)	Dielectric Constant
Natural		
Bamboo	6×10^{13}–7×10^{14}	5–7
Cork	1×10^{9}–1×10^{11}	6–8
Leather	1×10^{8}–1×10^{10}	5–10
Wood, typical (longitudinal)	6×10^{13}–2×10^{14}	5–6
Wood, typical (transverse)	2×10^{14}–7×10^{14}	5–6
Polymer		
Elastomer		
Butyl rubber	1×10^{15}–1×10^{16}	2.8–3.2
EVA	3.2×10^{21}–1×10^{22}	2.9–3.0
Isoprene (IR)	1×10^{15}–1×10^{16}	2.5–3.0
Natural rubber (NR)	1×10^{15}–1×10^{16}	3.0–4.5
Neoprene (CR)	1×10^{19}–1×10^{23}	6.7–8.0
Polyurethane elastomers	1×10^{18}–1×10^{22}	5.0–9.0
Silicone elastomers	3×10^{19}–1×10^{22}	2.9–4.0
Thermoplastic		
ABS	3×10^{21}–3×10^{22}	2.8–3.2
Cellulose polymers (ca)	1×10^{17}–4×10^{20}	3.0–5.0
Ionomer (I)	3×10^{21}–3×10^{22}	2.2–2.4
Nylons (PA)	1.5×10^{19}–1.1×10^{20}	3.7–3.9
Polycarbonate (PC)	1×10^{20}–1×10^{21}	3.1–3.3
PEEK	3×10^{21}–3×10^{22}	3.1–3.3
Polyethylene (PE)	3×10^{22}–3×10^{24}	2.2–2.4
PET	3×10^{20}–3×10^{21}	3.5–3.7
Acrylic (PMMA)	3×10^{23}–3×10^{24}	3.2–3.4
Acetal (POM)	3×10^{20}–3×10^{21}	3.6–4.0
Polypropylene (PP)	3×10^{22}–3×10^{23}	2.1–2.3
Polystyrene (PS)	1×10^{25}–1×10^{27}	3.0–3.3
Polyurethane thermoplastics	3×10^{18}–3×10^{19}	6.6–7.1
PVC	1×10^{20}–1×10^{22}	3.1–4.4
Teflon (PTFE)	3×10^{23}–3×10^{24}	2.1–2.2
Thermoset		
Epoxies	1×10^{20}–6×10^{21}	3.4–5.7
Phenolics	3×10^{18}–3×10^{19}	4.0–6.0
Polyester	3×10^{18}–3×10^{19}	2.8–3.3
Polymer foam		
Flexible polymer foam (VLD)	1×10^{20}–1×10^{23}	1.1–1.15
Flexible polymer foam (LD)	1×10^{20}–1×10^{23}	1.15–1.2
Flexible polymer foam (MD)	1×10^{20}–1×10^{23}	1.2–1.3
Rigid polymer foam (LD)	1×10^{17}–1×10^{21}	1.04–1.1
Rigid polymer foam (MD)	3×10^{16}–3×10^{20}	1.1–1.19
Rigid polymer foam (HD)	1×10^{16}–1×10^{20}	1.2–1.45

Table A.10 Embodied Energy and CO_2 Footprint

	Energy (MJ/kg)	CO_2 (kg/kg)
Polymer		
Elastomer		
Butyl rubber	95–120	3.6–4.2
EVA	87–96	2.9–3.2
Isoprene (IR)	77–85	2.2–2.4
Natural rubber (NR)	62–70	1.5–1.6
Neoprene (CR)	96–106	3.5–3.9
Polyurethane elastomers	109–120	4.5–4.9
Silicone elastomers	152–168	8.2–9.0
Thermoplastic		
ABS	91–102	3.3–3.6
Cellulose polymers (ca)	108–119	4.4–4.9
Ionomer (I)	102–112	4.0–4.4
Nylons (PA)	121–135	5.5–5.6
Polycarbonate (PC)	105–116	5.4–5.9
PEEK	223–246	13.0–14.0
Polyethylene (PE)	77–85	2.0–2.2
PET	80–88	2.2–2.5
Acrylic (PMMA)	94–110	3.4–3.8
Acetal (POM)	100–110	3.8–4.2
Polypropylene (PP)	85–105	2.6–2.8
Polystyrene (PS)	86–99	2.7–3.0
Polyurethane thermoplastics	113–125	4.6–5.3
PVC	68–95	2.2–2.6
Teflon (PTFE)	145–160	7.1–7.8
Thermoset		
Epoxies	105–130	4.2–4.6
Phenolics	86–95	2.8–3.1
Polyester	84–93	2.7–3.0
Polymer foam		
Flexible polymer foam (VLD)	103–115	4.0–4.8
Flexible polymer foam (LD)	103–115	4.0–4.8
Flexible polymer foam (MD)	103–115	4.0–4.8
Rigid polymer foam (LD)	105–110	3.5–4.0
Rigid polymer foam (MD)	105–110	3.5–4.0
Rigid polymer foam (HD)	105–110	3.5–4.0
Metal		
Ferrous		
Cast irons	16–18	0.97–1.1
High carbon steels	29–35	2.2–2.8
Medium carbon steels	29–35	2.2–2.8
Low carbon steels	29–35	2.2–2.8
Low alloy steels	32–38	2.2–2.8
Stainless steels	77–85	4.9–5.4

Table A.10 *continued*

	Energy (MJ/kg)	CO_2 (kg/kg)
Nonferrous		
Aluminum alloys	200–220	11.2–12.8
Copper alloys	68–74	4.9–5.6
Lead alloys	53–58	3.3–3.7
Magnesium alloys	360–400	22.4–24.8
Nickel alloys	127–140	7.9–9.2
Titanium alloys	600–740	7.9–11
Tungsten alloys	313–346	19.7–21.8
Zinc alloys	70–75	3.7–4.0
Ceramic		
Glass		
Borosilicate glass	24–26	1.3–1.4
Glass ceramic	36–40	2.0–2.2
Silica glass	30–33	1.6–1.8
Soda-lime glass	14–17	0.7–1.0
Porous		
Brick	2.2–3.5	0.2–0.23
Concrete, typical	1–1.3	0.13–0.15
Stone	5.5–6.4	0.3–0.34
Technical		
Alumina	50–55	2.7–3.0
Aluminum nitride	209–231	11.3–12.5
Boron carbide	153–169	8.3–9.1
Silicon (commercial grade)	56–62	3.8–4.2
Silicon carbide	70–78	6.3–6.9
Silicon nitride	116–128	3.1–3.4
Tungsten carbide	82–91	4.4–4.9
Composite		
Metal		
Aluminum/silicon carbide	250–300	7.5–8.3
Polymer		
CFRP	259–286	16.1–18.3
GFRP	107–118	7.5–8.3
Natural		
Bamboo	4–6	0.3–0.33
Cork	4–5	0.19–0.22
Leather	102–113	4.1–4.5
Wood, typical (longitudinal)	7–8	0.4–0.46
Wood, typical (transverse)	7–8	0.45–0.49

Table A.11 Approximate Material Prices, C_m

	C_m ($/kg)
Metal	
Ferrous	
Cast irons	0.57–0.69
High carbon steels	0.72–0.8
Medium carbon steels	0.67–0.74
Low carbon steels	0.63–0.7
Low alloy steels	0.81–0.89
Stainless steels	6.5–7.2
Nonferrous	
Aluminum alloys	1.5–1.7
Copper alloys	3.2–3.5
Lead alloys	3.2–3.5
Magnesium alloys	5.2–5.7
Nickel alloys	33–37
Titanium alloys	67–74
Zinc alloys	1.2–1.3
Ceramic	
Glass	
Borosilicate glass	4.1–6.2
Glass ceramic	2.1–12
Silica glass	6.2–10
Soda-lime glass	1.4–1.7
Porous	
Brick	0.62–1.7
Concrete, typical	0.041–0.062
Stone	0.4–0.6
Technical	
Alumina	18–27
Aluminum nitride	100–179
Boron carbide	60–89
Silicon	9.1–15
Silicon carbide	15–21
Silicon nitride	35–54
Tungsten carbide	19–29
Composite	
Metal	
Aluminum/silicon carbide	6.2–8.3
Polymer	
CFRP	40–44
GFRP	19–21

Table A.11 *continued*

	C_m ($/kg)
Polymer *continued*	
SMC	5.8–6.4
DMC	4.7–5.2
Natural	
Bamboo	1.4–2.1
Cork	2.8–14
Leather	17–21
Wood	0.8–1
Polymer	
Elastomer	
Butyl rubber	3.7–4.1
EVA	1.5–1.7
Isoprene (IR)	2.9–3.2
Natural rubber (NR)	1.7–1.9
Neoprene (CR)	5.1–5.6
Polyurethane elastomers (elPU)	4.9–5.3
Silicone elastomers	13–14
Thermoplastic	
ABS	2.1–2.5
Cellulose polymers (CA)	3.9–4.3
Ionomer (I)	2.8–3.7
Nylons (PA)	3.3–3.6
Polycarbonate (PC)	3.7–4.0
PEEK	100–120
Polyethylene (PE)	1.3–1.4
PET	1.5–1.7
Acrylic (PMMA)	2.6–2.8
Acetal (POM)	3.3–3.8
Polypropylene (PP)	1.2–1.3
Polystyrene (PS)	1.4–1.6
Polyurethane thermoplastics (tpPU)	4.1–5.6
PVC	0.93–1.0
Teflon (PTFE)	11–21
Thermoset	
Epoxies	2.6–2.8
Phenolics	1.7–1.9
Polyester	4.1–4.5
Polymer foam	
Flexible polymer foam (VLD)	2.9–3.1
Flexible polymer foam (LD)	2.9–3.1
Flexible polymer foam (MD)	2.9–3.1
Rigid polymer foam (LD)	12–25
Rigid polymer foam (MD)	12–25
Rigid polymer foam (HD)	12–25

WAYS OF CHECKING AND ESTIMATING DATA

The value of a database of material properties depends on its precision and its completeness—in short, on its quality. One way of maintaining or enhancing its quality is to subject its contents to validating procedures. The property range checks and dimensionless correlations, described below, provide powerful tools for doing this. The same procedures fill a second function: providing estimates for missing data, essential when no direct measurements are available.

Property ranges

Each property of a given class of materials has a characteristic *range*. A convenient way of presenting the information is as a table in which a low (L) and a high (H) value are stored, identified by the material family and class. An example listing Young's modulus, E, is shown in Table A.12, in which E_L is the lower limit and E_H the upper limit.

All properties have characteristic ranges like these. The range becomes narrower if the classes are made more restrictive. For purposes of checking and estimation, described in a moment, it is helpful to break down the family of *metals* into classes of cast irons, steels, aluminum alloys, magnesium alloys, titanium alloys, copper alloys, and so on. Similar subdivisions for polymers (thermoplastics, thermosets, elastomers) and for ceramics and

Table A.12 Ranges of Young's Modulus, E, for Broad Material Classes

Material class	E_L (GPa)	E_H (GPa)
All solids	0.00001	1,000
Classes of solid		
Metals: ferrous	70	220
Metals: nonferrous	4.6	570
Technical ceramics*	91	1,000
Glasses	47	83
Polymers: thermoplastic	0.1	4.1
Polymers: thermosets	2.5	10
Polymers: elastomers	0.0005	0.1
Polymeric foams	0.00001	2
Composites: metal-matrix	81	180
Composites: polymer-matrix	2.5	240
Woods: parallel to grain	1.8	34
Woods: perpendicular to grain	0.1	18

** Technical ceramics are dense, monolithic ceramics such as SiC, Al_2O_3, ZrO_2, and so on.*

glasses (engineering ceramics, whiteware, silicate glasses, minerals) increase resolution here also.

Correlations between material properties

Materials that are stiff have high melting points. Solids with low densities have high specific heats. Metals with high thermal conductivities have high electrical conductivities. These rules of thumb describe correlations between two or more material properties that can be expressed more quantitatively as limits for the values of *dimensionless property groups*. They take the form

$$C_L < P_1 P_2{}^n < C_H \tag{A.1}$$

or

$$C_L < P_1 P_2{}^n P_3{}^m < C_H \tag{A.2}$$

(or larger groupings), where P_1, P_2, P_3 are material properties, n and m are powers (usually −1, −1/2, +1/2, or +1), and C_L and C_H are dimensionless constants—the lower and upper limits between which the values of the property group lies. The correlations exert tight constraints on the data, giving the "patterns" of property envelopes that appear on the material selection charts. An example is the relationship between expansion coefficient, α (units: K^{-1}), and the melting point, T_m (units: K) or, for amorphous materials, the glass temperature, T_g:

$$C_L \le \alpha T_m \le C_H \tag{A.3a}$$

$$C_L \le \alpha T_g \le C_H \tag{A.3b}$$

a correlation with the form of Equation (A.1). Values for the dimensionless limits C_L and C_H for this group are listed in Table A.13 for a number of material classes. The values span a factor of 2 to 10 rather than the factor of 10 to 100 of the property ranges. There are many such correlations. They form the basis of a hierarchical scheme for data checking and estimating (one used in preparing the charts in this book), described next.

Data checking

Data checks proceed in three steps. Each datum is first associated with a material class or, at a higher level of checking, with a subclass. This identifies the values of the property range and correlation limits against which it will be checked. The datum is then compared with the range limits L and H for that class and property. If it lies within the range limits, it is accepted; if it does not, it is flagged for checking.

Table A.13 Limits for the Groups αT_m and αT_g for Broad Material Classes

Correlation* $C_L < \alpha T_m < C_H$	C_L ($\times 10^{-3}$)	C_H ($\times 10^{-3}$)
All solids	0.1	56
Classes of solid		
Metals: ferrous	13	27
Metals: nonferrous	2	21
Fine ceramics*	6	24
Glasses	0.3	3
Polymers: thermoplastic	18	35
Polymers: thermosets	11	41
Polymers: elastomers	35	56
Polymeric foams	16	37
Composites: metal-matrix	10	20
Composites: polymer-matrix	0.1	10
Woods: parallel to grain	2	4
Woods: perpendicular to grain	6	17

** For amorphous solids, the melting point, T_m, is replaced by the glass temperature, T_g.*

Why bother with such low-level stuff? Because it provides a sanity check. The commonest error in handbooks and other compilations of material or process properties is that of a value expressed in the wrong units or, for less obvious reasons, in error by one or more orders of magnitude (slipped decimal point, for instance). Range checks catch errors of this sort. If a demonstration of this is needed, it can be found by applying them to the contents of almost any standard reference data-book; none among those we have tried has passed without errors.

In the third step, each of the dimensionless groups of properties like the group in Table A.13 is formed in turn, and compared with the range bracketed by the limits C_L and C_H. If the value lies within its correlation limits, it is accepted; if not, it is checked. Correlation checks are more discerning than range checks and catch subtler errors, allowing the quality of the data to be further enhanced.

Data estimation

The relationships have another, equally useful, function. There remain gaps in our knowledge of material properties. The fracture toughness of many materials has not yet been measured, nor has the electric breakdown potential; even moduli are not always known. The absence of a datum for a material would falsely eliminate it from a screening exercise that used that property, even though the material might be a viable candidate. This difficulty is avoided by

using the correlation and range limits to estimate a value for the missing datum, adding a flag to alert the user that it is an estimate.

In estimating property values, the procedure used for checking is reversed: The dimensionless groups are used first because they are the more accurate. They can be surprisingly good. As an example, consider estimating the expansion coefficient, α, of polycarbonate from its glass temperature, T_g. Inverting Equation (A.3b) gives the estimation rule:

$$\frac{C_L}{T_g} \leq \alpha \leq \frac{C_H}{T_g} \tag{A.4}$$

Inserting values of C_L and C_H from Table A.13, and the value T_g = 420 K for a particular sample of polycarbonate gives the mean estimate

$$\overline{\alpha} = 63 \times 10^{-3}\,\mathrm{K}^{-1} \tag{A.5}$$

The reported value for polycarbonate is

$$\alpha = 54\text{–}62 \times 10^{-3}\,\mathrm{K}^{-1}$$

The estimate is within 9% of the mean of the measured values, perfectly adequate for screening purposes. That it is an estimate must not be forgotten, however: If thermal expansion is crucial to the design, better data or direct measurements are essential.

Only when the potential of the correlations is exhausted are the property ranges invoked. They provide a crude first estimate of the value of the missing property, far less accurate than the correlations but still useful in providing guide values for screening.

FURTHER READING

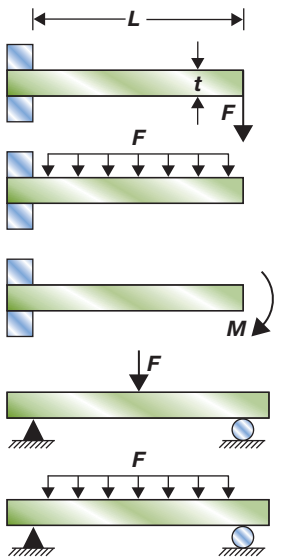
Ashby, M.F. (1998). Checks and estimates for material properties. Cambridge University Engineering Department. *Proc. Roy. Soc. A., 454*, 1301–1321.
This and the next reference detail the data-checking and data-estimating methods, listing the correlations.

Bassetti, D., Brechet, Y., & Ashby, M.F. (1998). Estimates for material properties: The method of multiple correlations. *Proc. Roy. Soc. A., 454*, 1323–1336.
This and the previous reference detail the data-checking and data estimating methods, with examples.

This page intentionally left blank

Appendix B: Useful Solutions for Standard Problems

Deflection of beams



C_1	C_2
3	2
8	6
2	1
48	16
$\frac{384}{5}$	24

$$\delta = \frac{FL^3}{C_1 EI} = \frac{ML^2}{C_1 EI}$$

$$\theta = \frac{FL^2}{C_2 EI} = \frac{ML}{C_2 EI}$$

Materials Selection in Mechanical Design. DOI: 10.1016/B978-1-85617-663-7.00019-9
© 2011 Michael F. Ashby. Published by Elsevier Ltd. All rights reserved.

CONTENTS

Introduction and Synopsis 527
B.1 Constitutive Equations for Mechanical Response 528
B.2 Moments of Sections 530
B.3 Elastic Bending of Beams 532
B.4 Failure of Beams and Panels 534
B.5 Buckling of Columns, Plates, and Shells 536
B.6 Torsion of Shafts 538
B.7 Static and Spinning Disks 540
B.8 Contact Stresses 542
B.9 Estimates for Stress Concentrations 544
B.10 Sharp Cracks 546
B.11 Pressure Vessels 548
B.12 Vibrating Beams, Tubes, and Disks 550
B.13 Creep and Creep Fracture 552
B.14 Flow of Heat and Matter 554
B.15 Solutions for Diffusion Equations 556
B.16 Further Reading 558

INTRODUCTION AND SYNOPSIS

Modeling is a key part of design. In the early stage, approximate modeling establishes whether the concept will work at all, and identifies the combination of material properties that maximize performance. At the embodiment stage, more accurate modeling brackets values for the forces, the displacements, the velocities, the heat fluxes, and the dimensions of the components. And in the final stage, modeling gives precise values for stresses, strains and failure probability in key components: power, speed, efficiency, and so on.

Many components with simple geometries and loads have been modeled already. Many more complex components can be modeled approximately by idealizing them as one of these. There is no need to reinvent the beam or the column or the pressure vessel; their behavior under all common types of loading has already been analyzed. The important thing is to know that the results exist and where to find them.

This appendix summarizes the results of modeling a number of standard problems. The usefulness of these results cannot be overstated. Many problems of conceptual design can be treated, with adequate precision, by patching together the solutions given here; even the detailed analysis of noncritical components can often be tackled in the same way. Even when this approximate approach is not sufficiently accurate, the insight it gives is valuable.

This appendix contains 15 double-page sections that list, with a short commentary, results for constitutive equations; for the loading of beams, columns, and torsion bars; for contact stresses, cracks, and other stress concentrations; for pressure vessels, vibrating beams, and plates; and for the flow of heat and matter. They are drawn from numerous sources, listed in Section B.16.

B.1 CONSTITUTIVE EQUATIONS FOR MECHANICAL RESPONSE

The behavior of a component when it is loaded depends on the *mechanism* by which it deforms. A beam loaded in bending may deflect elastically; it may yield plastically; it may deform by creep; and it may fracture in a brittle or ductile way. The equation that describes the material response is known as a *constitutive equation.* Each mechanism is characterized by a different constitutive equation, which contains one or more than one *material property*: Young's modulus, E, and Poisson's ratio, ν, are the material properties that enter the constitutive equation for linear-elastic deformation; the yield strength, σ_y, is the material property that enters the constitutive equation for plastic flow; creep constants, $\dot{\varepsilon}_o$, σ_o, and n enter the equation for creep; fracture toughness, K_{1c}, enters that for brittle fracture.

The common constitutive equations for mechanical deformation are listed on the facing page. In each case the equation for uniaxial loading by a tensile stress, σ is given first; below it is the equation for multiaxial loading by principal stresses σ_1, σ_2, and σ_3, always chosen so that σ_1 is the most tensile and σ_3 the most compressive (or least tensile). They are the basic equations that determine mechanical response.

Elastic

Uniaxial $\epsilon = \frac{\sigma}{E}$

General $\varepsilon_1 = \frac{\sigma_1}{E} - \nu(\sigma_2 + \sigma_3)$ etc.

Plastic

Uniaxial $\sigma \geq \sigma_y$

General $\sigma_1 - \sigma_3 \geq \sigma_y \quad (\sigma_1 > \sigma_2 > \sigma_3)$ or

$$\left[\frac{1}{2}\{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2\}\right]^{\frac{1}{2}} > \sigma_y$$

Creep

Uniaxial $\dot{\varepsilon} = \dot{\varepsilon}_o \left(\frac{\sigma}{\sigma_o}\right)^n$

General $\dot{\varepsilon}_1 = \dot{\varepsilon}_o \frac{\sigma_o^{n-1}}{\sigma_o^n} \{\sigma_1 - \frac{1}{2}(\sigma_2 + \sigma_3)\}$ etc.

Fracture

Uniaxial $\sigma \geq C K_{1c}/(\pi a)^{1/2}$

General $\sigma_1 \geq C K_{1c}/(\pi a)^{1/2} \quad (\sigma_1 > \sigma_2 > \sigma_3)$

E	= Young's modulus
n	= Poisson's ratio
σ_y	= Plastic yield strength
n, ε_o, σ_o	= Creep constants
K_{1C}, a	= Fracture toughness, crack length
C	≈ 1 Tension
C	≈ 15 Compression

FIGURE B.1

B.2 MOMENTS OF SECTIONS

A beam of uniform section, loaded in simple tension by a force, F, carries a stress

$$\sigma = F/A$$

where A is the area of the section. Its response is calculated from the appropriate constitutive equation. Here the important characteristic of the section is its area, A. For other modes of loading, higher moments of the area are involved. Those for various common sections are given on the facing page. They are defined as follows.

The second moment, I, measures the resistance of the section to bending about a horizontal axis (shown as a broken line). It is

$$I = \int_{\text{section}} y^2 \, b(y) \, dy$$

where y is measured vertically and $b(y)$ is the width of the section at y. The moment, K, measures the resistance of the section to twisting. It is equal to the polar moment J for circular sections, where

$$J = \int_{\text{section}} 2\pi r^3 dr$$

where r is measured radially from the center of the circular section. For noncircular sections, K is less than J.

The section modulus $Z = I/y_m$ (where y_m is the normal distance from the neutral axis of bending to the outer surface of the beam) measures the surface stress generated by a given bending moment, M:

$$\sigma = \frac{My_m}{I} = \frac{M}{Z}$$

Finally, the moment Z_p, defined by

$$Z_p = \int_{\text{section}} y \, b(y) \, dy$$

measures the resistance of the beam to fully plastic bending. The fully plastic moment for a beam in bending is

$$M_p = Z_p \sigma_y$$

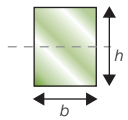
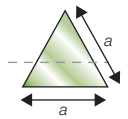
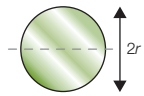
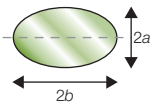
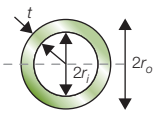
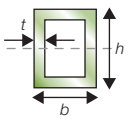
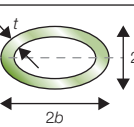
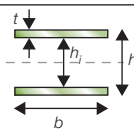
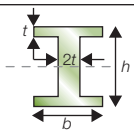
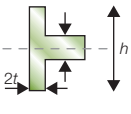
Section Shape	Area A m^2	Moment I m^4	Moment K m^4	Moment Z m^3	Moment Z_p m^3
	bh	$\frac{bh^3}{12}$	$\frac{bh^3}{3}(1-0.58\frac{b}{h})$ $(h>b)$	$\frac{bh^2}{6}$	$\frac{bh^2}{4}$
	$\frac{\sqrt{3}}{4}a^2$	$\frac{a^4}{32\sqrt{3}}$	$\frac{\sqrt{3}a^4}{80}$	$\frac{a^3}{32}$	$\frac{3a^3}{64}$
	πr^2	$\frac{\pi}{4}r^4$	$\frac{\pi}{2}r^4$	$\frac{\pi}{4}r^3$	$\frac{\pi}{3}r^3$
	πab	$\frac{\pi}{4}a^3b$	$\frac{\pi a^3b^3}{(a^2+b^2)}$	$\frac{\pi}{4}a^2b$	$\frac{\pi}{3}a^2b$
	$\pi(r_o^2-r_i^2)$ $\approx 2\pi rt$	$\frac{\pi}{4}(r_o^4-r_i^4)$ $\approx \pi r^3 t$	$\frac{\pi}{2}(r_o^4-r_i^4)$ $\approx 2\pi r^3 t$	$\frac{\pi}{4r_o}(r_o^4-r_i^4)$ $\approx \pi r^2 t$	$\frac{\pi}{3}(r_o^3-r_i^3)$ $\approx \pi r^2 t$
	$2t(h+b)$ $(h, b >> t)$	$\frac{1}{6}h^3t(1+3\frac{b}{h})$	$\frac{2tb^2h^2}{(h+b)}(1-\frac{t}{h})^4$	$\frac{1}{3}h^2t(1+3\frac{b}{h})$	$bht(1+\frac{h}{2b})$
	$\pi(a+b)t$ $(a, b >> t)$	$\frac{\pi}{4}a^3t(1+\frac{3b}{a})$	$\frac{4\pi(ab)^{5/2}t}{(a^2+b^2)}$	$\frac{\pi}{4}a^2t(1+\frac{3b}{a})$	$\pi abt(2+\frac{a}{b})$
	$b(h_o-h_i)$ $\approx 2bt$ $(h, b >> t)$	$\frac{b}{12}(h_o^3-h_i^3)$ $\approx \frac{1}{2}bth_o^2$	–	$\frac{b}{6h_o}(h_o^3-h_i^3)$ $\approx bth_o$	$\frac{b}{4}(h_o^2-h_i^2)$ $\approx bth_o$
	$2t(h+b)$ $(h, b >> t)$	$\frac{1}{6}h^3t(1+3\frac{b}{h})$	$\frac{2}{3}bt^3(1+4\frac{h}{b})$	$\frac{1}{3}h^2t(1+3\frac{b}{h})$	$bht(1+\frac{h}{2b})$
	$2t(h+b)$ $(h, b >> t)$	$\frac{t}{6}(h^3+4bt^2)$	$\frac{t^3}{3}(8b+h)$	$\frac{t}{3h}(h^3+4bt^2)$	$\frac{th^2}{2}\{1+\frac{2t(b-2t)}{h^2}\}$

FIGURE B.2

B.3 ELASTIC BENDING OF BEAMS

When a beam is loaded by a force, F, or moments, M, the initially straight axis is deformed into a curve. If the beam is uniform in section and properties, long in relation to its depth, and nowhere stressed beyond the elastic limit, the deflection, δ, and the angle of rotation, θ, can be calculated using elastic beam theory (see Section B.16). The basic differential equation describing the curvature of the beam at a point x along its length is

$$EI\frac{d^2y}{dx^2} = M$$

where y is the lateral deflection, and M is the bending moment at the point x on the beam. E is Young's modulus, and I is the second moment of area (Section B.2). When M is constant this becomes

$$\frac{M}{I} = E\left(\frac{1}{R} - \frac{1}{R_o}\right)$$

where R_o is the radius of curvature before applying the moment and R is the radius after it is applied. Deflections δ and rotations θ are found by integrating these equations along the beam.

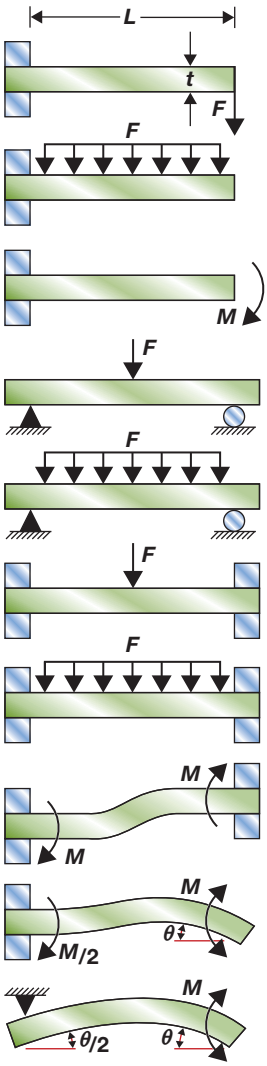
The stiffness of the beam is defined by

$$S = \frac{F}{\delta} = \frac{C_1 EI}{L^3}$$

It depends on Young's modulus, E, for the material of the beam, on its length, L, and on the second moment of its section, I. The end slope of the beam, θ, is given by

$$\theta = \frac{FL^2}{C_2 EI}$$

Equations for the deflection, δ, and end slope, θ, of beams for various common modes of loading are shown on the facing page together with values of C_1 and C_2.



C_1	C_2
3	2
8	6
2	1
48	16
$\frac{384}{5}$	24
192	–
384	–
6	–
–	4
–	3

$$\delta = \frac{FL^3}{C_1EI} = \frac{ML^2}{C_1EI}$$

$$\theta = \frac{FL^2}{C_2EI} = \frac{ML}{C_2EI}$$

E = Young's modulus (N/m^2)
δ = Deflection (m)
F = Force (N)
M = Moment (Nm)
L = Length (m)
b = Width (m)
t = Depth (m)
θ = End slope (–)
I = See Table B.2 (m^4)
y = Distance from neutral axis (m)
R = Radius of curvature (m)

$$\frac{\sigma}{y} = \frac{M}{I} = \frac{E}{R}$$

FIGURE B.3

B.4 FAILURE OF BEAMS AND PANELS

The longitudinal (or "fiber") stress, σ, at a point, y, from the neutral axis of a uniform beam loaded elastically in bending by a moment, M, is

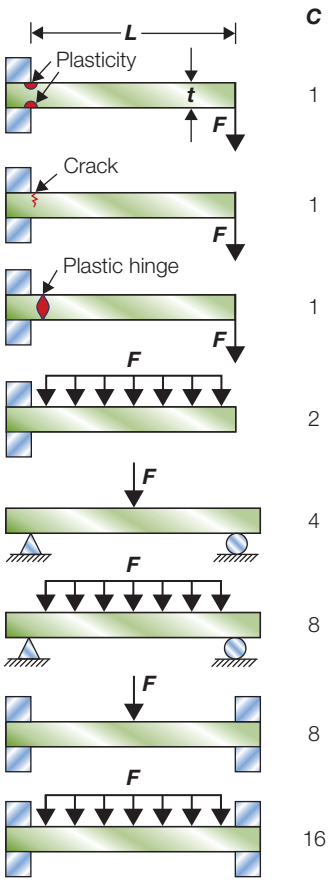
$$\frac{\sigma}{y} = \frac{M}{I} = E\left(\frac{1}{R} - \frac{1}{R_o}\right)$$

where I is the second moment of area (Section B.2), E is Young's modulus, Ro is the radius of curvature before applying the moment, and R is the radius after it is applied. The tensile stress in the outer fiber of such a beam is

$$\sigma = \frac{My_m}{I} = \frac{M}{Z}$$

where y_m is the perpendicular distance from the neutral axis to the outer surface of the beam. If this stress reaches the yield strength, σ_y, of the material of the beam, small zones of plasticity appear at the surface (*top*, facing page). The beam is no longer elastic and, in this sense, has failed. If, instead, the maximum fiber stress reaches the brittle fracture strength, σ_f (the "modulus of rupture," often shortened to *MOR*) of the material of the beam, a crack nucleates at the surface and propagates inward (*middle*); in this case, the beam has certainly failed. A third criterion for failure is often important: that the plastic zones penetrate through the section of the beam, linking to form a plastic hinge (*bottom*).

The failure moments and failure loads, for each of these three types of failure, and for each of several geometries of loading, are given in the diagram. The formulae labeled "onset" refer to the first two failure modes; those labeled "full plasticity" refer to the third. Two new functions of section shape are involved. Onset of failure involves the quantity Z; full plasticity involves the quantity H. Both are listed in the table in Section B.2, and defined in the text that accompanies it.



	C
Plasticity	1
Crack	1
Plastic hinge	1
	2
	4
	8
	8
	16

$$M_f = Z\sigma^* \quad \text{(onset)}$$

$$M_f = Z_p\sigma_y \quad \text{(full plasticity)}$$

$$F_f = CZ\frac{\sigma^*}{L} \quad \text{(onset)}$$

$$F_f = \frac{CZ_p\sigma_y}{L} \quad \text{(full plasticity)}$$

M_f = Failure moment (Nm)

F_f = Force at failure (N)

L = Length (m)

t = Depth (m)

b = Width (m)

I = See Table B.2 (m^4)

Z = See Table B.2 (m^3)

Z_p = See Table B.2 (m^3)

σ_y = Yield strength (N/m^2)

σ_f = Modulus of rupture (N/m^2)

$\sigma^* = \sigma_y$ (plastic material)

$= \sigma_f$ (brittle material)

FIGURE B.4

B.5 BUCKLING OF COLUMNS, PLATES, AND SHELLS

If sufficiently slender, an elastic column loaded in compression fails by elastic buckling at a critical load, F_{crit}. This load is determined by the end constraints, of which four extreme cases are illustrated on the facing page: An end may be constrained in a position and direction; it may be free to rotate but not to translate (or "sway"); it may sway without rotation; and it may both sway and rotate. Pairs of these constraints applied to the ends of the column lead to the five cases shown on the facing page. Each is characterized by a value of the constant n that is equal to the number of half-wavelengths of the buckled shape.

The addition of the bending moment, M, reduces the buckling load by the amount shown in the second box. A negative value of F_{crit} means that a tensile force is necessary to prevent buckling.

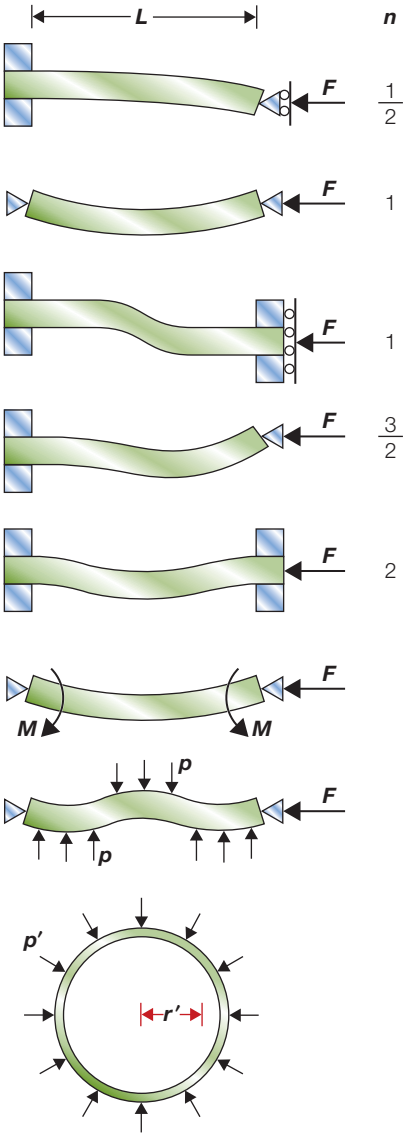
An elastic foundation is one that exerts a lateral restoring pressure, p, proportional to the deflection:

$$p = ky$$

where k is the foundation stiffness per unit depth and y is the local lateral deflection. Its effect is to increase F_{crit}, by the amount shown in the third box.

A thin-walled elastic tube will buckle inward under an external pressure, p', given in the last box. Here I refers to the second moment of area of a section of the tube wall cut parallel to the tube axis.

Thin or slender shapes may buckle locally before they yield or fracture. It is this that sets a practical limit to the thinness of tube walls and webs (Chapter 9).



$$F_{crit} = \frac{n^2\pi^2 EI}{L}$$

or $$\frac{F_{crit}}{A} = \frac{n^2\pi^2 E}{(L/r)^2}$$

F = Force (N)
M = Moment (Nm)
E = Young's modulus (N/m^3)
L = Length (m)
A = Section area (m^2)
I = See Table B.2 (m^4)
r = Gyration radius $(I/A)^{1/2}$ (m)
k = Foundation stiffness (N/m^2)
n = Half-wavelengths in buckled shape
p' = Pressure (N/m^2)

$$F_{crit} = \frac{\pi^2 EI}{L^2} - \frac{M^2}{4EI}$$

$$F_{crit} = \frac{n^2\pi^2 EI}{L^2} - \frac{kL^2}{n^2}$$

$$p'_{crit} = \frac{3EI}{(r')^3}$$

FIGURE B.5

B.6 TORSION OF SHAFTS

A torque, T, applied to the ends of an isotropic bar of uniform section and acting in the plane normal to the axis of the bar produces an angle of twist, θ. The twist is related to the torque by the first equation on the facing page, in which G is the shear modulus. For round bars and tubes of circular section, the factor K is equal to J, the polar moment of inertia of the section, defined in Section B.2. For any other section shape, K is less than J. Values of K are given in Section B.2.

If the bar ceases to deform elastically, it is said to have failed. This will happen if the maximum surface stress exceeds either the yield strength, σ_y, of the material or the stress at which it fractures, σ_{fr}. For circular sections, the shear stress at any point a distance, r, from the axis of rotation is

$$\tau = \frac{Tr}{K} = \frac{G\theta r}{K}$$

The maximum shear stress, τ_{max}, and the maximum tensile stress, σ_{max}, are at the surface and have the values

$$\tau_{max} = \sigma_{max} = \frac{Td_o}{2K} = \frac{G\theta d_o}{2L}$$

If τ_{max} exceeds $\sigma_y/2$ (using a Tresca yield criterion), or if σ_{max} exceeds σ_{fr}, the bar fails, as shown in the figure. The maximum surface stress for the solid ellipsoidal, square, rectangular, and triangular sections is at the points on the surface closest to the centroid of the section (the midpoints of the longer sides). It can be estimated approximately by inscribing the largest circle that can be contained within the section and calculating the surface stress for a circular bar of that diameter. More complex section shapes require special consideration and, if thin, may additionally fail by buckling.

Helical springs are a special case of torsional deformation. The extension of a helical spring of n turns of radius R, under a force, F, and the failure force, F_{crit}, are given on the facing page.

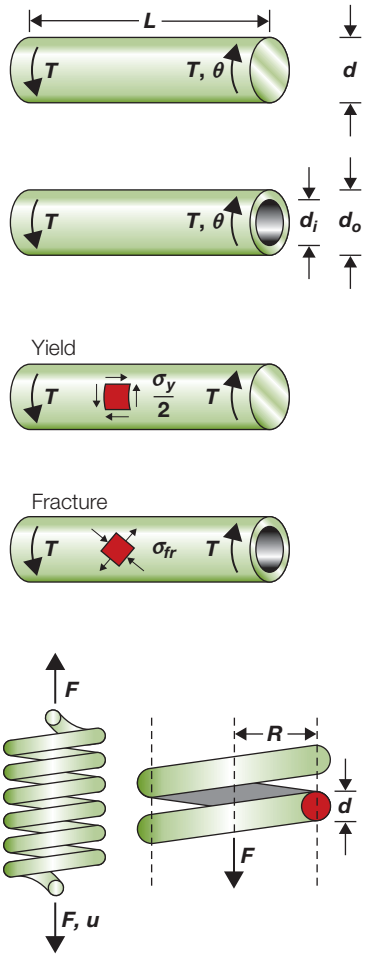


FIGURE B.6

Elastic deflection

$$\theta = \frac{LT}{KG}$$

Failure

$$T_f = \frac{K\sigma_y}{d_o} \text{ (onset of yield)}$$

$$T_f = \frac{2K\sigma_f}{d_o} \text{ (brittle fracture)}$$

T = Torque (Nm)
θ = Angle of twist
G = Shear modulus (N/m^2)
L = Length (m)
d = Diameter (m)
K = See Table B.2 (m^4)
σ_y = Yield strength (N/m^2)
σ_f = Modulus of rupture (N/m^2)

Spring deflection and failure

$$u = \frac{64FR^3n}{Gd^4}$$

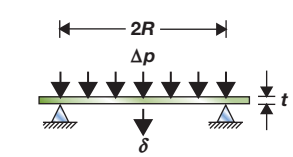
$$F_f = \frac{\pi}{32}\frac{d^3\sigma_y}{R}$$

F = Force (N)
u = Deflection (m)
R = Coil radius (m)
n = Number of turns

B.7 STATIC AND SPINNING DISKS

A thin disk deflects when a pressure difference, Δp, is applied across its two surfaces. The deflection causes stresses to appear in the disk. The first box on the facing page gives deflection and maximum stress (important in predicting failure) when the edges of the disk are simply supported. The second gives the same quantities when the edges are clamped. The results for a thin horizontal disk deflecting under its own weight are found by replacing Δp by the mass-per-unit area, $\rho\ g\ t$, of the disk (here ρ is the density of the material of the disk and g is the acceleration due to gravity). Thick disks are more complicated; for those, see "Further reading" (Section B.16).

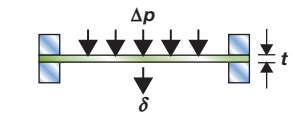
Spinning disks, rings, and cylinders store kinetic energy. Centrifugal forces generate stresses in the disk. The two boxes list the kinetic energy and the maximum stress, σ_{max}, in disks and rings rotating at an angular velocity ω (radians/sec). The maximum rotation rate and energy are limited by the burst strength of the disk. They are found by equating the maximum stress in the disk to the strength of the material.



Simply supported

$$\delta = \frac{3}{4}(1-\nu^2)\frac{\Delta p R^4}{Et^3}$$

$$\sigma_{max} = \frac{3}{8}(3+\nu)\frac{\Delta p R^2}{t^2}$$

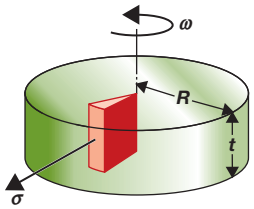


Clamped

$$\delta = \frac{3}{16}(1-\nu^2)\frac{\Delta p R^4}{Et^3}$$

$$\sigma_{max} = \frac{3}{8}(1+\nu)\frac{\Delta p R^2}{t^2}$$

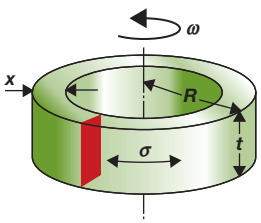
δ = Deflection (m)
E = Young's modulus (N/m^2)
Δp = Pressure difference (N/m^2)
ν = Poisson's ratio



Disk

$$U = \frac{\pi}{4}\rho t \omega^2 R^4$$

$$\sigma_{max} = \frac{1}{8}(3+\nu)\rho\omega^2 R^2$$



Ring

$$U = \pi\rho t \omega^2 R^3 x$$

$$\sigma_{max} = \rho\omega^2 R^2$$

U = Energy (J)
ω = Angular velocity (rad/s)
ρ = Density (kg/m^3)

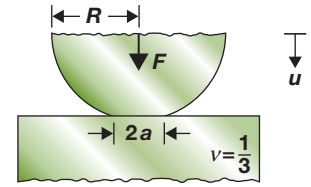
FIGURE B.7

B.8 CONTACT STRESSES

When surfaces are placed in contact, they touch at one or a few discrete points. If the surfaces are loaded, the contacts flatten elastically and the contact areas grow until failure of some sort occurs: failure by crushing (caused by the compressive stress, σ_c), tensile fracture (caused by the tensile stress, σ_t), or yielding (caused by the shear stress, σ_s). The boxes on the facing page summarize the important results for the radius, a, of the contact zone, the center-to-center displacement, u, and the peak values of σ_c, σ_t, and σ_s.

The first box shows results for a sphere on a flat, when both have the same moduli and Poisson's ratio has the value 1/3. Results for the more general problem (the "Hertzian indentation" problem) are shown in the second box: Two elastic spheres (radii R_1 and R_2, moduli and Poisson's ratios E_1, ν_1 and E_2, ν_2) are pressed together by a force, F.

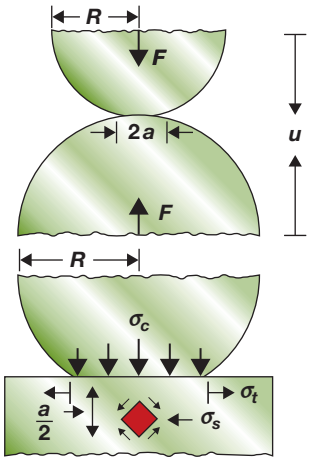
If the shear stress, σ_s, exceeds the shear yield strength, $\sigma_y/2$, a plastic zone appears beneath the center of the contact at a depth of about $a/2$ and spreads to form the fully plastic field shown in the two lower figures. When this state is reached, the contact pressure is approximately three times the yield stress, as shown in the bottom box.



$$a = 0.7\left(\frac{FR}{E}\right)^{\frac{1}{3}}$$

$$u = 1.0\left(\frac{F^2}{E^2R}\right)^{\frac{1}{3}}$$

$\nu = 0.33$



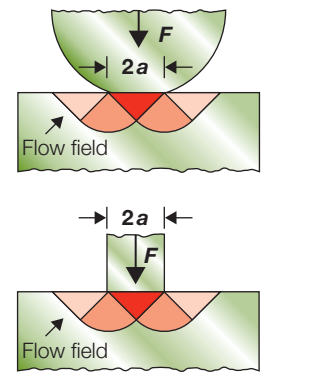
$$a = \left(\frac{3}{4}\frac{F}{E^*}\frac{R_1R_2}{(R_1+R_2)}\right)^{\frac{1}{3}}$$

$$u = \left(\frac{9}{16}\frac{F^2}{(E^*)^2}\frac{(R_1+R_2)}{R_1R_2}\right)^{\frac{1}{3}}$$

$$(\sigma_c)_{max} = \frac{3F}{2\pi a^2}$$

$$(\sigma_s)_{max} = \frac{F}{2\pi a^2}$$

$$(\sigma_t)_{max} = \frac{F}{6\pi a^2}$$



R_1, R = Radii of spheres (m)

E_1, E_2 = Moduli of spheres (N/m^2)

ν_1, ν_2 = Poisson's ratio

F = Load (N)

a = Radius of contact (m)

u = Displacement (m)

σ = Stresses (N/m^2)

σ_y = Yield stress (N/m^2)

$$E^* = \left(\frac{1-\nu_1^2}{E_1} + \frac{1-\nu_2^2}{E_2}\right)^{-1}$$

$$\frac{F}{\pi a^2} = 3\sigma_y$$

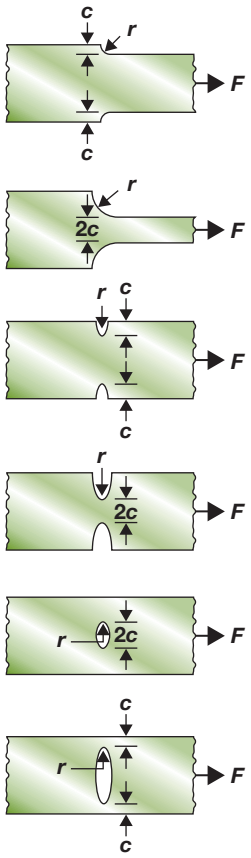
FIGURE B.8

B.9 ESTIMATES FOR STRESS CONCENTRATIONS

Stresses and strains are concentrated at holes, slots, or changes of section in elastic bodies. Plastic flow, fracture, and fatigue cracking start at these places. The local stresses at the stress concentrations can be computed numerically, but this is often unnecessary. Instead, they can be estimated using the equation shown on the facing page.

The stress concentration caused by a change in section dies away at distances on the order of the characteristic dimension of the section change (defined more fully below), an example of St Venant's principle at work. This means that the maximum local stresses in a structure can be found by determining the nominal stress distribution, neglecting local discontinuities (such as holes or grooves), and then multiplying the nominal stress by a stress concentration factor. Elastic stress concentration factors are given approximately by the equation. In it, σ_{nom} is defined as the load divided by the minimum cross-section of the part, ρ is the minimum radius of curvature of the stress-concentrating groove or hole, and c is the characteristic dimension: either the half-thickness of the remaining ligament, the half-length of a contained crack, the length of an edge-crack, or the height of a shoulder, whichever is *least*. The drawings show examples of each such situation. The factor α is roughly 2 for tension, but is nearer 1/2 for torsion and bending. Though inexact, the equation is an adequate working approximation for many design problems.

The maximum stress is limited by plastic flow or fracture. When plastic flow starts, the strain concentration grows rapidly while the stress concentration remains constant. The strain concentration becomes the more important quantity, and may not die out rapidly with distance (St Venant's principle no longer applies).



$$\frac{\sigma_{max}}{\sigma_{nom}} = 1 + \alpha\left(\frac{c}{r}\right)^{\frac{1}{2}}$$

F = Force (N)

A_{min} = Minimum section (m^2)

$\sigma_{nom} = F/A_{min}$ (N/m^2)

r = Radius of curvature (m)

c = Characteristic length (m)

α ≈ 0.5 (torsion)

α ≈ 2.0 (tension)

FIGURE B.9

B.10 SHARP CRACKS

Sharp cracks (that is, stress concentrations with a tip radius of curvature of atomic dimensions) concentrate stress in an elastic body more acutely than rounded stress concentrations do. To a first approximation, the local stress falls off as $1/r^{1/2}$ with radial distance r from the crack tip. A tensile stress, σ, applied normal to the plane of a crack of length $2a$ contained in an infinite plate (as in the top figure on the facing page) gives rise to a local stress field, σ_ℓ, that is tensile in the plane containing the crack and given by

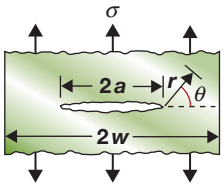
$$\sigma_\ell = \frac{C\sigma\sqrt{\pi a}}{\sqrt{2\pi r}}$$

where r is measured from the crack tip in the plane $\theta = 0$ and C is a constant. The mode 1 *stress intensity factor*, K_I, is defined as

$$K_1 = C\sigma\sqrt{\pi a}$$

Values of the constant C for various modes of loading are given in the figure. The stress σ for point loads and moments is given by the equations at the bottom. The crack propagates when $K_1 > K_{1c}$, the *fracture toughness*.

When the crack length is very small compared with all specimen dimensions and compared with the distance over which the applied stress varies, C is equal to 1 for a contained crack and 1.1 for an edge crack. As the crack extends in a uniformly loaded component, it interacts with the free surfaces, giving the correction factors shown opposite. If, in addition, the stress field is nonuniform (as it is in an elastically bent beam), C differs from 1; two examples are given in the figure. The factors, C, given here, are approximate only, good when the crack is short but not when the crack tips are very close to the boundaries of the sample. They are adequate for most design calculations. More accurate approximations and other less common loading geometries can be found in the references listed in "Further reading" (Section B.16).

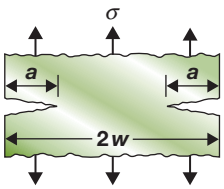


C

1.0 if $a \ll w$

$$\left(\frac{2w}{\pi a}\tan\frac{\pi a}{2w}\right)^{\frac{1}{2}}$$

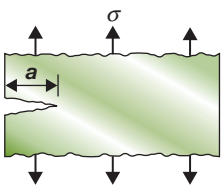
$K_1 = C\sigma\sqrt{\pi a}$
Failure when
$K_1 \geq K_{1c}$



1.1 if $a \ll w$

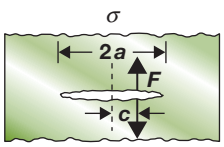
$$\left(\frac{\frac{2w}{\pi a}\tan\frac{\pi a}{2w}}{1-\frac{a}{w}}\right)^{\frac{1}{2}}$$

K_1 = Stress intensity (N/m$^{\frac{3}{2}}$)
σ = Remote stress (N/m^2)
F = Load (N)
M = Moment (Nm)
a = Internal crack half-length (m)
= Surface crack length (m)
w = Half-width (center) (m)
= Width (edge crack) (m)
b = Sample depth (m)
t = Beam thickness (m)



1.1 if $a \ll w$

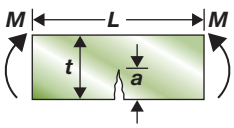
$$\frac{1.1\left(1-0.2\frac{a}{w}\right)}{\left(1-\frac{a}{w}\right)^{\frac{3}{2}}}$$



$$\frac{2}{\pi}\left(\frac{a+c}{a-c}\right)^{\frac{1}{2}}$$

Point load on crack face

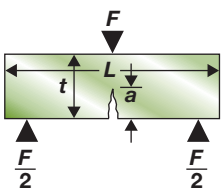
$$\sigma = \frac{F}{2ab}$$



$$\frac{1.1\left(1-\frac{3}{2}\frac{a}{t}\right)}{\left(1-\frac{a}{t}\right)^{\frac{3}{2}}}$$

Moment on beam

$$\sigma = \frac{6M}{bt}$$



$$\frac{1.1\left(1-\frac{3}{2}\frac{a}{t}\right)}{\left(1-\frac{a}{t}\right)^{\frac{3}{2}}}$$

3-point bending

$$\sigma = \frac{3FL}{2bt}$$

FIGURE B.10

B.11 PRESSURE VESSELS

Thin-walled pressure vessels are treated as membranes. The approximation is reasonable when $t < b/4$. The stresses in the wall are given on the facing page; they do not vary significantly with radial distance, r. Those in the plane tangent to the skin, σ_θ and σ_z for the cylinder and σ_θ and σ_φ for the sphere, are just equal to the internal pressure amplified by the ratio b/t or $b/2t$, depending on geometry. The radial stress, σ_r, is equal to the mean of the internal and external stress, $p/2$ in this case. The equations describe the stresses when an external pressure, p_e, is superimposed if p is replaced by $(p - p_e)$.

In thick-walled vessels, the stresses vary with radial distance, r, from the inner to the outer surfaces, and are greatest at the inner surface. The equations can be adapted for the case of both internal and external pressures by noting that when the internal and external pressures are equal, the state of stress in the wall is

$$\sigma_\theta = \sigma_r = -p \text{ (cylinder)}$$

or

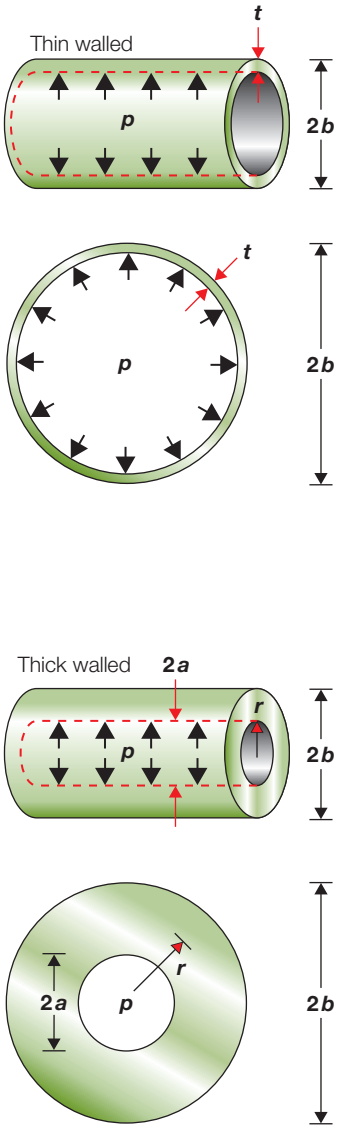
$$\sigma_\theta = \sigma_\phi = \sigma_r = -p \text{ (sphere)}$$

allowing the term involving the external pressure to be evaluated. It is not valid to just replace p by $(p - p_e)$.

Pressure vessels fail by yielding when the von Mises equivalent stress first exceeds the yield strength, σ_y. They fail by fracture if the largest tensile stress exceeds the fracture stress, σ_{fr}, where

$$\sigma_{fr} = \frac{CK_{1c}}{\sqrt{\pi a}}$$

and K_{1c} is the fracture toughness, a is the half-crack length, and C is a constant given in Section B.10.



Cylinder

$$\sigma_\theta = \frac{pb}{t}$$

$$\sigma_r = -\frac{p}{2}$$

$$\sigma_z = \frac{pb}{2t} \quad \text{(Closed ends)}$$

Sphere

$$\sigma_\theta = \sigma_\phi = \frac{pb}{2t}$$

$$\sigma_r = -\frac{p}{2}$$

p = Pressure (N/m^2)

t = Wall thickness (m)

a = Inner radius (m)

b = Outer radius (m)

r = Radial coordinate (m)

Cylinder

$$\sigma_\theta = \frac{pa^2}{r^2}\left(\frac{b^2 - r^2}{b^2 - a^2}\right)$$

$$\sigma_r = -\frac{pa^2}{r^2}\left(\frac{b^2 + r^2}{b^2 - a^2}\right)$$

Sphere

$$\sigma_\theta = \sigma_\phi = \frac{pa^3}{2r^3}\left(\frac{b^3 + 2r^3}{b^3 - a^3}\right)$$

$$\sigma_r = -\frac{pa^3}{r^3}\left(\frac{b^3 - r^3}{b^3 - a^3}\right)$$

FIGURE B.11

B.12 VIBRATING BEAMS, TUBES, AND DISKS

Any undamped system vibrating at one of its natural frequencies can be reduced to the simple problem of a mass, m, attached to a spring of stiffness K. The lowest natural frequency of such a system is

$$f = \frac{1}{2\pi}\sqrt{\frac{K}{m}}$$

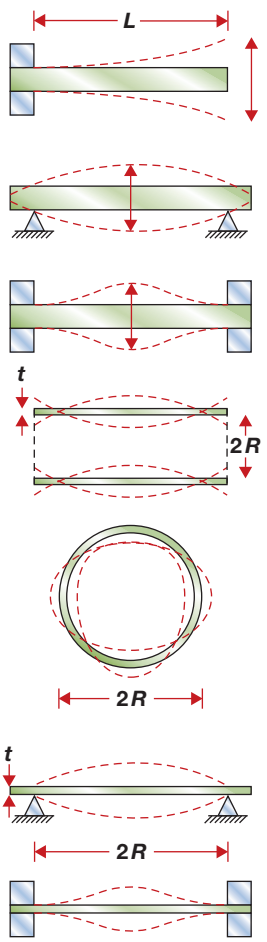
Specific cases require specific values for m and K. They can often be estimated with sufficient accuracy to be useful in approximate modeling. Higher natural frequencies are simple multiples of the lowest.

The first box on the facing page gives the lowest natural frequencies of the flexural modes of uniform beams with various end constraints. As an example, the first can be estimated by assuming that the effective mass of the beam is one quarter of its real mass, so that

$$m = \frac{m_o L}{4}$$

where m_o is the mass per unit length of the beam, and K is the bending stiffness (given by F/δ from Section B.3); the estimate differs from the exact value by 2%. Vibrations of a tube have a similar form, using I and m_o for the tube. Circumferential vibrations can be found approximately by "unwrapping" the tube and treating it as a vibrating plate, simply supported at two of its four edges.

The second box gives the lowest natural frequencies for flat circular disks with simply supported and clamped edges. Disks with doubly curved faces are stiffer and have higher natural frequencies.



	C_1	
		Beams and tubes
	3.52	$f = \frac{C_1}{2\pi}\sqrt{\frac{EI}{m_o L^4}}$
	9.87	f = Natural frequency (s^{-1}) $m_o = \rho A$ = Mass/length (kg/m) ρ = Density (kg/m^3) A = Section area (m^2) I = See Table B.2
	22.4	
	9.87	With $A = 2\pi Rt$ $I = \pi R^3 t$
	2.68	With $A = Lt$ $I = \frac{Lt^3}{12}$ L = Length of tube
	C_2	Disks
	1.44	$f = \frac{C_2}{2\pi}\sqrt{\frac{Et^3}{m_1 R^4(1-\nu^2)}}$
	2.94	$m_1 = \rho t$ = Mass/area (kg/m^2) t = Thickness (m) R = Radius (m) ν = Poisson's ratio

FIGURE B.12

B.13 CREEP AND CREEP FRACTURE

At temperatures above 1/3 T_m (where T_m is the absolute melting point), materials creep when loaded. It is convenient to characterize the creep of a material by its behavior under a tensile stress, σ, at a temperature T_m. Under these conditions the steady-state tensile strain rate, $\dot{\varepsilon}_{ss}$, is often found to vary as a power of the stress and exponentially with temperature:

$$\dot{\varepsilon}_{ss} = A\left(\frac{\sigma}{\sigma_o}\right)^n exp - \frac{Q}{RT}$$

where Q is an activation energy, A is a kinetic constant, and R is the gas constant. At constant temperature this becomes

$$\dot{\varepsilon}_{ss} = \dot{\varepsilon}_o(\frac{\sigma}{\sigma_o})n$$

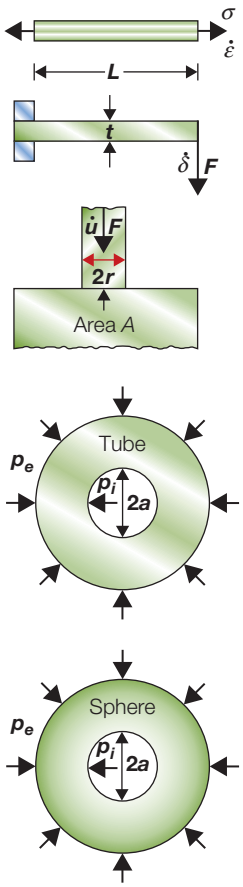
where $\dot{\varepsilon}_o$ (s^{-1}), $\sigma_o(N/m^2)$, and n are creep constants.

The behavior of creeping components is summarized on the facing page, which give the deflection rate of a beam, the displacement rate of an indenter and the change in relative density of cylindrical and spherical pressure vessels in terms of the tensile creep constants.

Prolonged creep causes the accumulation of creep damage that ultimately leads, after a time t_f, to fracture. To a useful approximation

$$t_f\dot{\varepsilon}_{ss} = C$$

where C is a constant characteristic of the material. Creep-ductile materials have values of C between 0.1 and 0.5; creep-brittle materials have values of C as low as 0.01.



$$\dot{\varepsilon} = \dot{\varepsilon}_o \left(\frac{\sigma}{\sigma_o}\right)^n$$

$$\dot{\delta} = \frac{2\dot{\varepsilon}_o}{n+2}\left(\frac{(2n+1)}{n s_o}\frac{F}{bt}\right)^n \left(\frac{L}{t}\right)^{n+1} L$$

$$\dot{u} = C_1 \dot{\varepsilon}_o \sqrt{A}\left(\frac{C_2 F}{\sigma_o A}\right)^n$$

σ = Stress (N/m^2)
F = Force (N)
$\dot{\delta}, \dot{u}$ = Displacement rates (m/s)
n, $\dot{\varepsilon}_o$, σ_o = Creep constants
L, b, t = Beam dimensions (m)
a, b = Radii of pressure vessels (m)
ρ = Relative density, $\frac{b^3 - a^3}{b^3}$
$\dot{\rho}$ = Rate of change of ρ
C_1, C_2 = Constants

$$\dot{\rho} = 2\dot{\varepsilon}_o \frac{\rho(1-\rho)}{(1-(1-\rho)^{1/n})^n}\left(\frac{2}{n}\frac{(p_e - p_i)}{\sigma_o}\right)^n$$

$$\dot{\rho} = \frac{3}{2}\dot{\varepsilon}_o \frac{\rho(1-\rho)}{(1-(1-\rho)^{1/n})^n}\left(\frac{3}{2n}\frac{(p_e - p_i)}{\sigma_o}\right)^n$$

FIGURE B.13

B.14 FLOW OF HEAT AND MATTER

Heat flow can be limited by conduction, by convection, or by radiation. The constitutive equations for each are listed on the facing page. The first equation is Fourier's first law, describing steady-state heat flow; it contains the thermal conductivity, λ. The second is Fourier's second law, which treats transient heat-flow problems; it contains the thermal diffusivity, a, defined by

$$a = \frac{\lambda}{\rho C}$$

where ρ is the density and C is the specific heat at constant pressure. Solutions to these two differential equations are given in Section B.15.

The third equation describes convective heat transfer. It, rather than conduction, limits heat flow when the Biot number is

$$B_i = \frac{hs}{\lambda} < 1$$

where h is the heat-transfer coefficient and s is a characteristic dimension of the sample. When, instead, $B_i > 1$, heat flow is limited by conduction. The final equation is the Stefan-Boltzmann law for radiative heat transfer. The emissivity, ε, is unity for black bodies; less for all other surfaces.

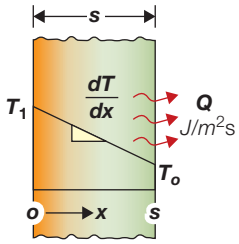
Diffusion of matter follows a pair of differential equations with the same form as Fourier's two laws, and with similar solutions. They are commonly written

$$J = -D\nabla C = -D\frac{dC}{dx} \text{ (steady state)}$$

and

$$\frac{\partial C}{\partial t} = D\nabla^2 C = D\frac{\partial^2 C}{\partial x^2} \text{ (time-dependent flow)}$$

where J is the flux, C is the concentration, x is the distance, and t is time. Solutions are given in Section B.15.



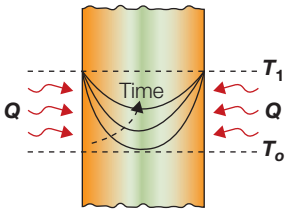
$$Q = -\lambda \nabla T = -\lambda \frac{dT}{dx}$$

Q = Heat flux (J/m^{2}s)

T = Temperature (K)

x = Distance (m)

λ = Thermal conductivity (W/mK)



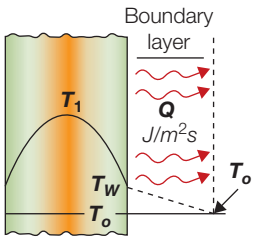
$$\frac{\delta T}{\delta t} = a \nabla^2 T = a \frac{\delta^2 T}{\delta x^2}$$

t = Time (s)

ρ = Density (kg/m^3)

C_p = Specific heat (J/kg.K)

a = Thermal diffusivity, $\frac{\lambda}{\rho Cp}$ (m^2/s)



$$Q = h(T_w - T_o)$$

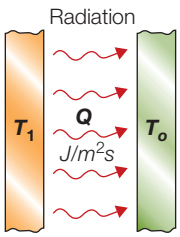
T_w = Surface temperature (K)

T = Fluid temperature (K)

h = Heat transfer coefficient (W/m^{2}K)

= 5 – 50 W/m^2.K in air

= 1,000 – 5,000 W/m^2.K in water



$$Q = \varepsilon \sigma (T_1^4 - T_o^4)$$

ε = Emissivity (1 for black body)

σ = Stefan's constant

= 5.67×10^{-8} W/m^2.K^4

FIGURE B.14

B.15 SOLUTIONS FOR DIFFUSION EQUATIONS

Solutions exist for the diffusion equations for a number of standard geometries. They are worth knowing because many real problems can be approximated by one of these.

At steady state the temperature or concentration profile does not change with time. This is expressed by the boxed equations within the first box at the top of the facing page. Solutions for these are given below for uniaxial flow, radial flow in a cylinder, and radial flow in a sphere. The solutions are fitted to individual cases by matching the constants A and B to the boundary conditions. Solutions for matter flow are found by replacing temperature, T, by concentration, C, and replacing conductivity, λ, by diffusion coefficient, D.

The box within the second large box summarizes the governing equations for time-dependent flow, assuming that the diffusivity (a or D) is not a function of position. Solutions for the temperature or concentration profiles, $T(x,t)$ or $C(x,t)$, are given below. The first equation gives the "thin-film" solution: A thin slab at temperature T_1 or concentration C_1 is sandwiched between two semi-infinite blocks at T_o or C_o, at $t = 0$, and flow allowed. The second result is for two semi-infinite blocks, initially at T_1 and T_o (or C_1 or C_o), brought together at $t = 0$. The last is for a T or C profile that is sinusoidal, of wavelength $\lambda/2\pi$ and amplitude A at $t = 0$.

Note that all transient problems end up with a characteristic time constant, t^*, with

$$t^* = \frac{x^2}{\beta a} \quad or \quad \frac{x^2}{\beta D}$$

where x is a dimension of the specimen; or they end up with a characteristic length, x^*, with

$$x^* = \sqrt{\beta a t} \quad or \quad \sqrt{\beta D t}$$

where t is the time scale of observation, with $1 < \beta < 4$, depending on geometry.

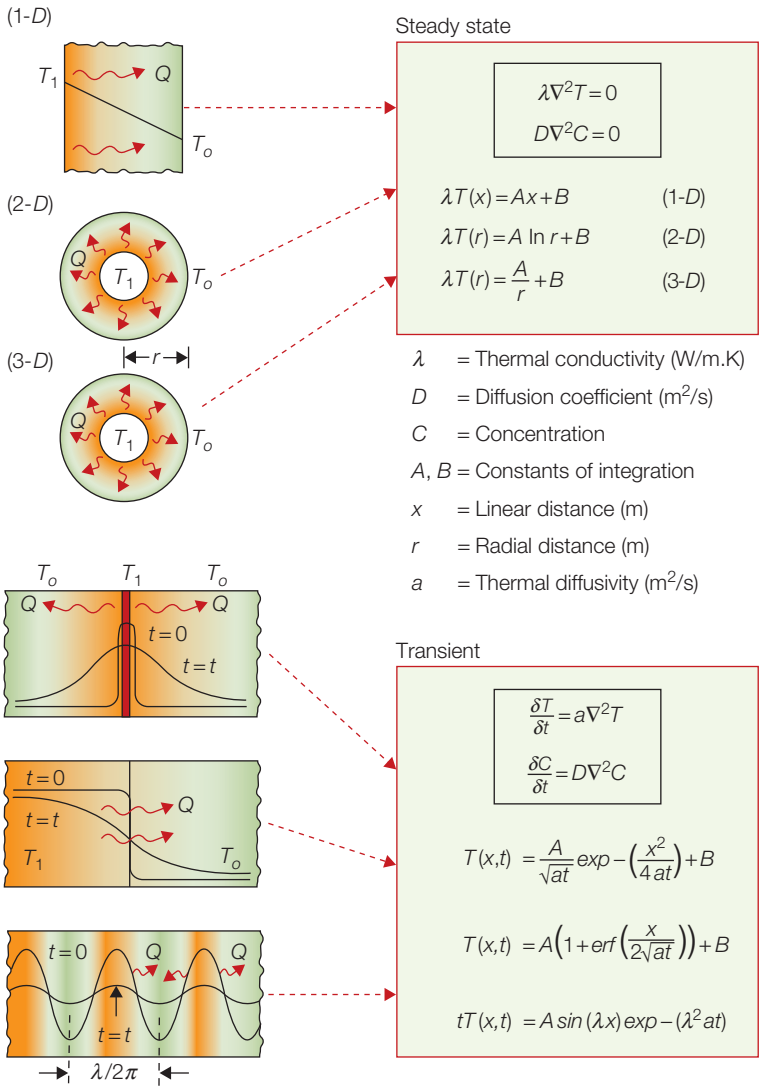

(1-D)
T_1
Q
T_o
(2-D)
Q
T_1
T_o
r
(3-D)
Q
T_1
T_o
Steady state
$\lambda \nabla^2 T = 0$
$D \nabla^2 C = 0$
$\lambda T(x) = Ax + B$ (1-D)
$\lambda T(r) = A \ln r + B$ (2-D)
$\lambda T(r) = \frac{A}{r} + B$ (3-D)
λ = Thermal conductivity (W/m.K)
D = Diffusion coefficient (m^2/s)
C = Concentration
A, B = Constants of integration
x = Linear distance (m)
r = Radial distance (m)
a = Thermal diffusivity (m^2/s)
T_o T_1 T_o
Q Q
t = 0
t = t
t = 0
t = t
Q
T_1
T_o
t = 0
Q Q
t = t
$\lambda/2\pi$
Transient
$\frac{\delta T}{\delta t} = a \nabla^2 T$
$\frac{\delta C}{\delta t} = D \nabla^2 C$
$T(x,t) = \frac{A}{\sqrt{at}} exp - \left(\frac{x^2}{4at}\right) + B$
$T(x,t) = A\left(1 + erf\left(\frac{x}{2\sqrt{at}}\right)\right) + B$
$tT(x,t) = A \sin(\lambda x) exp - (\lambda^2 at)$

FIGURE B.15

B.16 FURTHER READING

Constitutive laws

Cottrell, A.H. (1964). *Mechanical properties of matter*. Wiley.

Gere, J.M., & Timoshenko, S.P. (1985). *Mechanics of materials* (2nd SI ed.). Wadsworth International.

Moments of area

Young, W.C. (1989). *Roark's formulas for stress and strain* (6th ed.). McGraw-Hill.

Beams, shafts, columns, and shells

Calladine, C.R. (1983). *Theory of shell structures*. Cambridge University Press.

Gere, J.M., & Timoshenko, S.P. (1985). *Mechanics of materials* (2nd ed.). Wadsworth International.

Timoshenko, S.P., & Goodier, J.N. (1970). *Theory of elasticity* (3rd ed.). McGraw-Hill.

Timoshenko, S.P., & Gere, J.M. (1961). *Theory of elastic stability* (2nd ed.). McGraw-Hill.

Young, W.C. (1989). *Roark's formulas for stress and strain* (6th ed.). McGraw-Hill.

Contact stresses and stress concentration

Timoshenko, S.P., & Goodier, J.N. (1970). *Theory of elasticity* (3rd ed.). McGraw-Hill.

Hill, R. (1950). *Plasticity*. Oxford University Press.

Johnson, K.L. (1985). *Contact mechanics*. Oxford University Press.

Sharp cracks

Hertzberg, R.W. (1989). *Deformation and fracture of engineering materials* (3rd ed.). Wiley.

Tada, H., Paris, P.C., & Irwin, G.R. (1985). *The stress analysis of cracks handbook* (2nd ed.). Paris Productions and Del Research Group.

Pressure vessels

Timoshenko, S.P., & Goodier, J.N. (1970). *Theory of elasticity* (3rd ed.). McGraw-Hill.

Hill, R. (1950). *Plasticity*. Oxford University Press.

Young, W.C. (1989). *Roark's formulas for stress and strain* (6th ed.). McGraw-Hill.

Vibration

Young, W.C. (1989). *Roark's formulas for stress and strain* (6th ed.). McGraw-Hill.

Creep

Finnie, I., & Heller, W.R. (1976). *Creep of engineering materials*. McGraw-Hill.

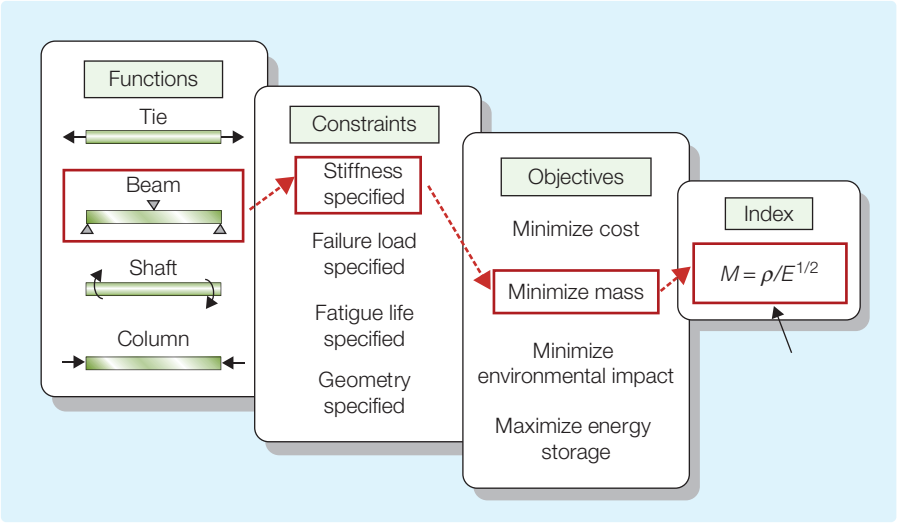
Heat and matter flow

Hollman, J.P. (1981). *Heat transfer* (5th ed.). McGraw-Hill.

Carslaw, H.S., & Jaeger, J.C. (1959). *Conduction of heat in solids* (2nd ed.). Oxford University Press.

Shewmon, P.G. (1989). *Diffusion in solids* (2nd ed.). TMS.

Appendix C: Material Indices



CONTENTS

C.1 Introduction and Synopsis 560

C.2 Uses of Material Indices 560

Material selection *560*

Material deployment or substitution *560*

How to read the tables *560*

The symbols *560*

Materials Selection in Mechanical Design. DOI: 10.1016/B978-1-85617-663-7.00020-5
© 2011 Michael F. Ashby. Published by Elsevier Ltd. All rights reserved.

C.1 INTRODUCTION AND SYNOPSIS

The performance, P, of a component is characterized by a performance equation. The performance equation contains groups of material properties. These groups are the material indices. Sometimes the "group" is a single property; thus if the performance of a beam is measured by its stiffness, the performance equation contains only one property, the elastic modulus, E. It is the material index for this problem. More commonly the performance equation contains a group of two or more properties. Familiar examples are the specific stiffness, E/ρ, and the specific strength, σ_y/ρ (where σ_y is the yield strength or elastic limit, and ρ is the density), but there are many others. They help to determine the optimal selection of materials. Details of the method, with numerous examples, are given in Chapters 5 through 8. This appendix compiles indices for a range of common applications.

C.2 USES OF MATERIAL INDICES

Material selection

Components have functions: to carry loads safely, to transmit heat, to store energy, to insulate, and so forth. Each function has an associated material index. Materials with high values of the appropriate index maximize that aspect of the performance of the component. For reasons given in Chapter 5, the material index is generally independent of the details of the design. Thus the indices for beams in the tables that follow are independent of the detailed shape of the beam; that for minimizing thermal distortion of precision instruments is independent of the configuration of the instrument, and so forth. This gives them great generality.

Material deployment or substitution

A new material may have potential application in functions for which its indices have unusually high or low values. Fruitful applications for a new material can be identified by evaluating its indices and comparing them with those of existing, established materials. Similar reasoning points the way to identifying viable substitutes for an incumbent material in an established application.

How to read the tables

The indices listed in Tables C.1 through C.7 are, for the most part, based on the objective of minimizing mass. To minimize material cost, use the index for minimum mass, replacing the density, ρ, with the cost per unit volume, $C_m\rho$, where C_m is the cost per kg. To minimize material-embodied energy or CO_2 burden, replace ρ with $H_p a \cdot \rho$ or by $CO_2 \cdot \rho$, where H_p is the production energy per kg and CO_2 is the CO_2 burden per kg (Table A.10).

The symbols

The symbols used in Tables C.1 through C.7 are defined in the following table.

Class	Property	Symbol and Units
General	Density	ρ (kg/m^3)
	Price	C_m ($/kg)
Mechanical	Elastic moduli (Young's, shear, bulk)	E, G, K (GPa)
	Poisson's ratio	ν (–)
	Failure strength (yield, fracture)	σ_f (MPa)
	Fatigue strength	σ_e (MPa)
	Hardness	H (Vickers)
	Fracture toughness	K_{1c} (MPa.m$^{1/2}$)
	Loss coefficient (damping capacity)	η (–)
Thermal	Thermal conductivity	λ (W/m.K)
	Thermal diffusivity	a (m^2/s)
	Specific heat	C_p (J/kg.K)
	Thermal expansion coefficient	α (°K^{-1})
	Difference in thermal conductivity	$\Delta\alpha$ (°K^{-1})
Electrical	Electrical resistivity	ρ_e ($\mu\Omega$.cm)
Eco-properties	Energy/kg to produce material per kg CO_2/kg burden of material production per kg	E_p (MJ/kg) CO_2 (kg/kg)

Table C.1 Stiffness-limited Design at Minimum Mass

Function and Constraints	Maximize
TIE (tensile strut)	
Stiffness, length specified; section area free	E/ρ
SHAFT (loaded in torsion)	
Stiffness, length, shape specified; section area free	$G^{1/2}/\rho$
Stiffness, length, outer radius specified; wall thickness free	G/ρ
Stiffness, length, wall thickness specified; outer radius free	$G^{1/3}/\rho$
BEAM (loaded in bending)	
Stiffness, length, shape specified; section area free	$E^{1/2}/\rho$
Stiffness, length, height specified; width free	E/ρ
Stiffness, length, width specified; height free	$E^{1/3}/\rho$
COLUMN (compression strut, failure by elastic buckling)	
Buckling load, length, shape specified; section area free	$E^{1/2}/\rho$
PANEL (flat plate, loaded in bending)	
Stiffness, length, width specified; thickness free	$E^{1/3}/\rho$
PLATE (flat plate, compressed in-plane, buckling failure)	
Collapse load, length, width specified; thickness free	$E^{1/3}/\rho$
CYLINDER WITH INTERNAL PRESSURE	
Elastic distortion, pressure, radius specified; wall thickness free	E/ρ
SPHERICAL SHELL WITH INTERNAL PRESSURE	
Elastic distortion, pressure, radius specified; wall thickness free	$E/(1-\nu)\rho$

Table C.2 Strength-limited Design at Minimum Mass

Function and Constraints	Maximize
TIE (tensile strut)	
Stiffness, length specified; section area free	σ_f/ρ
SHAFT (loaded in torsion)	
Load, length, shape specified; section area free	$\sigma_f^{2/3}/\rho$
Load, length, outer radius specified; wall thickness free	σ_f/ρ
Load, length, wall thickness specified; outer radius free	$\sigma_f^{1/2}/\rho$
BEAM (loaded in bending)	
Load, length, shape specified; section area free	$\sigma_f^{2/3}/\rho$
Load length, height specified; width free	σ_f/ρ
Load, length, width specified; height free	$\sigma_f^{1/2}/\rho$
COLUMN (compression strut)	
Load, length, shape specified; section area free	σ_f/ρ
PANEL (flat plate, loaded in bending)	
Stiffness, length, width specified; thickness free	$\sigma_f^{1/2}/\rho$
PLATE (flat plate, compressed in-plane, buckling failure)	
Collapse load, length, width specified; thickness free	$\sigma_f^{1/2}/\rho$
CYLINDER WITH INTERNAL PRESSURE	
Elastic distortion, pressure, radius specified; wall thickness free	σ_f/ρ
SPHERICAL SHELL WITH INTERNAL PRESSURE	
Elastic distortion, pressure, radius specified; wall thickness free	σ_f/ρ
FLYWHEELS, ROTATING DISKS	
Maximum energy storage per unit volume; given velocity	ρ
Maximum energy storage per unit mass; no failure	σ_f/ρ

Table C.3 Strength-limited Design: Springs and Hinges

Function and Constraints	Maximize
SPRINGS	
Maximum stored elastic energy per unit volume; no failure	σ_f^2/E
Maximum stored elastic energy per unit mass; no failure	$\sigma_f^2/E\rho$
ELASTIC HINGES	
Radius of bend to be minimized (max flexibility without failure)	σ_f/E
KNIFE EDGES, PIVOTS	
Minimum contact area, maximum bearing load	σ_f^3/E^2 and H
COMPRESSION SEALS AND GASKETS	
Maximum conformability; limit on contact pressure	$\sigma_f^{3/2}/E$ and $1/E$
DIAPHRAGMS	
Maximum deflection under specified pressure or force	$\sigma_f^{3/2}/E$
ROTATING DRUMS AND CENTRIFUGES	
Maximum angular velocity; radius fixed; wall thickness free	σ_f/ρ

Table C.4 Vibration-limited Design

Function and Constraints	Maximize
TIES, COLUMNS	
Maximum longitudinal vibration frequencies	E/ρ
BEAMS, all dimensions prescribed	
Maximum flexural vibration frequencies	E/ρ
BEAMS, length and stiffness prescribed	
Maximum flexural vibration frequencies	$E^{1/2}/\rho$
PANELS, all dimensions prescribed	
Maximum flexural vibration frequencies	E/ρ
PANELS, length, width, stiffness prescribed	
Maximum flexural vibration frequencies	$E^{1/3}/\rho$
TIES, COLUMNS, BEAMS, PANELS (stiffness prescribed)	
Minimum longitudinal excitation from external drivers, ties	$\eta E/\rho$
Minimum flexural excitation from external drivers, beams	$\eta E^{1/2}/\rho$
Minimum flexural excitation from external drivers, panels	$\eta E^{1/3}/\rho$

Table C.5 Damage-tolerant Design

Function and Constraints	Maximize
TIES (tensile member)	
Maximum flaw tolerance and strength, load-controlled design	K_{1c} and σ_f
Maximum flaw tolerance and strength, displacement control	K_{1c}/E and σ_f
Maximum flaw tolerance and strength, energy control	K_{1c}^2/E and σ_f
SHAFTS (loaded in torsion)	
Maximum flaw tolerance and strength, load-controlled design	K_{1c} and σ_f
Maximum flaw tolerance and strength, displacement control	K_{1c}/E and σ_f
Maximum flaw tolerance and strength, energy control	K_{1c}^2/E and σ_f
BEAMS (loaded in bending)	
Maximum flaw tolerance and strength, load-controlled design	K_{1c} and σ_f
Maximum flaw tolerance and strength, displacement control	K_{1c}/E and σ_f
Maximum flaw tolerance and strength, energy control	K_{1c}^2/E and σ_f
PRESSURE VESSEL	
Yield-before-break	K_{1c}/σ_f
Leak-before-break	K_{1c}^2/σ_f

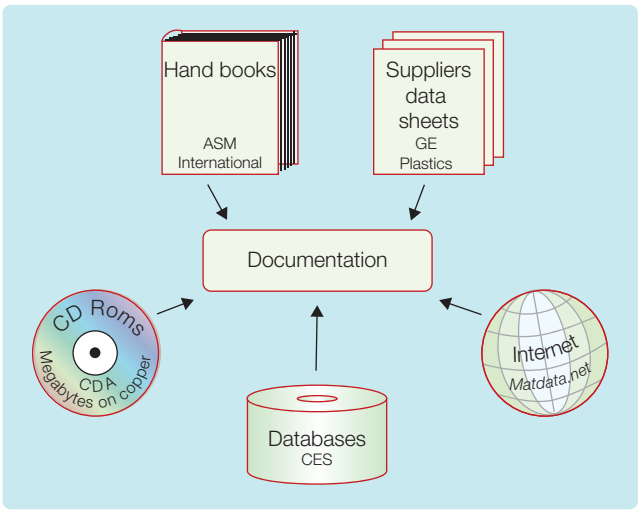
Table C.6 Electro-mechanical Design

Function and Constraints	Maximize
BUS BARS	
Minimum life cost; high current conductor	$1/\rho_e \rho C_m$
ELECTRO-MAGNET WINDINGS	
Maximum short-pulse field; no mechanical failure	σ_f
Maximum field and pulse length; limit on temperature rise	$C_p\rho/\rho_e$
WINDINGS, HIGH-SPEED ELECTRIC MOTORS	
Maximum rotational speed; no fatigue failure	σ_e/ρ_e
Minimum ohmic losses; no fatigue failure	$1/\rho_e$
RELAY ARMS	
Minimum response time; no fatigue failure	$\sigma_e/E\rho_e$
Minimum ohmic losses; no fatigue failure	$\sigma_e^2/E\rho_e$

Table C.7 Thermal and Thermo-mechanical Design

Function and Constraints	Maximize
THERMAL INSULATION MATERIALS	
Minimum heat flux at steady state; thickness specified	$1/\lambda$
Minimum temp rise in specified time; thickness specified	$1/a = \rho C_p/\lambda$
Minimize total energy consumed in thermal cycle (kilns, etc.)	$\sqrt{a}/\lambda = \sqrt{1/\lambda\rho C_p}$
THERMAL STORAGE MATERIALS	
Maximum energy stored/unit material cost (storage heaters)	C_p/C_m
Maximize energy stored for given temperature rise and time	$\lambda/\sqrt{a} = \sqrt{\lambda\rho C_p}$
PRECISION DEVICES	
Minimize thermal distortion for given heat flux	λ/α
THERMAL SHOCK RESISTANCE	
Maximum change in surface temperature; no failure	$\sigma_f/E\alpha$
HEAT SINKS	
Maximum heat flux per unit volume; expansion limited	$\lambda/\Delta\alpha$
Maximum heat flux per unit mass; expansion limited	$\lambda/\rho\Delta\alpha$
HEAT EXCHANGERS (pressure-limited)	
Maximum heat flux per unit area; no failure under Δp	$\lambda\sigma_f$
Maximum heat flux per unit mass; no failure under Δp	$\lambda\sigma_f/\rho$

Appendix D: Data Sources for Documentation



Materials Selection in Mechanical Design. DOI: 10.1016/B978-1-85617-663-7.00021-7
© 2011 Michael F. Ashby. Published by Elsevier Ltd. All rights reserved.

CONTENTS

D.1 Introduction 566
D.2 Information Sources for Materials 566
D.3 Information for Manufacturing Processes 589
D.4 Databases and Expert Systems as Software 590
D.5 Additional Useful Internet Sites 592

D.1 INTRODUCTION

This appendix tells you where to look to find information, both structured and unstructured, for material attributes. The sources, broadly speaking, are of three sorts: hard copy, software, and the Internet. The hard-copy documents listed below will be found in most engineering libraries. The computer databases are harder to find: The supplier is listed, along with the most recent website address. Internet sites are easy to find but can be frustrating to use.

Section D.2 catalogs information sources for families and classes of material, with a brief commentary where appropriate. Section D.3 provides a starting point for reading about processes. Section D.4 lists software for materials and process data, information, and selection. Section D.5 lists additional Internet sites on which materials information such as price can be found.

D.2 INFORMATION SOURCES FOR MATERIALS

D2.1 All materials

Few hard-copy data sources span the full spectrum of materials and properties. Ten that, in different ways, attempt to do so are listed here.

Materials Selector (1997). *Materials Engineering, Special Issue.* Penton Publishing.
Tabular data for a broad range of metals, ceramics, polymers, and composites.

Waterman, N.A., & Ashby, M.F. (Eds.) (1996). *The Chapman and Hall materials selector*. Chapman and Hall.
A three-volume compilation of data for all materials, with selection and design guide. Basic reference work.

ASM handbook series (1980–2010). ASM International.
More than 20 volumes of information about materials, emphasizing metals. Now available online.

Bauccio, M.L. (Ed.) (1994). *ASM engineered materials reference book* (2nd ed.). ASM International.
Compact compilation of numeric data for metals, polymers, ceramics, and composites.

Materials selector and design guide (1974). Design Engineering, Morgan-Grampian Ltd.
Resembles the Materials Engineering Materials Selector, *but less detailed and now rather dated.*

Handbook of industrial materials (2nd ed.) (1992). Elsevier.
A compilation of data remarkable for its breadth: metals, ceramics, polymers, composites, fibers, sandwich structures, leather …

Brady, G.S., & Clauser, H.R. (Eds.) (1986). *Materials handbook* (12th ed.). McGraw-Hill.
A broad survey, covering metals, ceramics, polymers, composites, fibers, sandwich structures, and more.

Cardarelli, F. (1999). *Materials handbook*. Springer-Verlag.
A remarkable compilation of material data spanning mechanical, thermal, and electrical properties of all material classes.

Goldsmith, A., Waterman, T.E., & Hirschhorn, J.J. (1961). *Handbook of thermophysical properties of solid materials*. MacMillan.
Thermophysical and thermochemical data for elements and compounds.

Bittence, J.C. (Ed.) (1994). *Guide to engineering materials producers*. ASM International.
A comprehensive catalog of addresses for material suppliers.

Internet sources of information on all classes of materials

ASM Handbooks online, *www.asminternational.org/hbk/index.jsp*

ASM Alloy center, *www.asminternational.org/alloycenter/index.jsp*

ASM Materials Information, *www.asminternational.org/matinfo/index.jsp*

A to Z of Materials, *www.azom.com*

Design InSite, *www.designinsite.dk*

Goodfellow, *www.goodfellow.com*

K&K Associate's thermal connection, *www.tak2000.com*

Corrosion Source (databases), *www.corrosionsource.com*

Material Data Network, *www.matdata.net*

Materials (Research): Alfa Aesar, *www.alfa.com*

MatWeb, *www.matweb.com*

MSC datamart, *www.mscsoftware.com*

NASA Long Duration Exposure Facility, SETAS, *http://setas-www.larc.nasa.gov/LDEF/*

NPL MIDAS, *midas.npl.co.uk/midas/index.jsp*

D2.2 All metals

Metals and alloys conform to national and (sometimes) international standards. One consequence is the high quality of data. Hard-copy sources for metals data are generally comprehensive, well-structured, and easy to use.

ASM metals handbook (9th and 10th eds.) (1986, 1990). ASM International.
Basic reference work, continuously upgraded and expanded. The 10th edition contains Vol. 1: Irons and steels; Vol. 2: Non-ferrous alloys; Vol. 3: Heat treatment; Vol. 4: Friction, lubrication, and wear; Vol. 5: Surface finishing and coating; Vol. 6: Welding and brazing; Vol. 7: Microstructural analysis. *More volumes have been released.*

Bauccio, M.L. (Ed.) (1993). *ASM metals reference book* (3rd ed.). ASM International.
Consolidates data for metals from a number of ASM publications. Basic reference work.

Smithells, C.J. (1992). *Metals reference book* (7th ed.). E.A. Brandes & G.B. Brook (Eds.). Butterworths.
A comprehensive compilation of data for metals and alloys. Basic reference work.

Robb, C. (1990). *Metals databook*. The Institute of Metals.
A concise collection of data on metallic materials covered by U.K. specifications only.

ASM guide to materials engineering data and information (1986). ASM International.
A directory of suppliers, trade organizations, and publications on metals.

Bringas, J.E. (Ed.) (1992). *The metals black book, vol. 1, steels*. Casti Publishing.
A compact book of data for steels.

Bringas, J.E. (Ed.) (1993). *The metals red book, vol. 2, nonferrous metals*. Casti Publishing.

Internet sources for two or more metals classes

ASM International Handbooks, *www.asminternational.org/hbk/index.jsp*

ASM International, Alloy Center, *www.asminternational.org/alloycenter/index.jsp*

Carpenter Technology home page, *www.cartech.com*

CASTI Publishing site catalog, *www.casti-publishing.com/intsite.htm*

CMW Inc. home page, *www.cmwinc.com*

Eurometaux, European Association of Metals, *www.eurometaux.org*

Materials (high performance): MatTech, *www.mat-tech.com*

Metsteel.com, *www.metsteel.com*

Rare earths: Pacific Industrial Development Corp., *www.pidc.com*

Rare metals: Stanford Materials Inc., *www.stanfordmaterials.com*

Refractory metals: Teledyne Wah Chang, *www.twca.com*

D2.3 Nonferrous metals and alloys

In addition to the references listed in Section D2.2, the following sources give data for specific metals and alloys

Pure metals

Most of the sources listed in the previous section contain some information on pure metals. However, the publiations listed below are particularly useful in this respect.

Emsley, J. (1989). *The elements*. Oxford University Press.
A book aimed more at chemists and physicists than engineers, with good coverage of chemical, thermal, and electrical properties but not mechanical properties.

Brandes, E.A., & Brook, G.B. (Eds.) (1992). *Smithells metals reference book* (7th edition), Butterworth-Heinemann.
Data for the mechanical, thermal, and electrical properties of pure metals.

Goodfellow catalogue (1995–96). Goodfellow Cambridge Limited.
Useful though patchy data for mechanical, thermal, and electrical properties of pure metals in a tabular format. Free.

Alfa Aesar catalog (1995–96). Johnson Matthey Catalog Co. Inc.
Coverage similar to that of the Goodfellow catalogue. Free.

Samsonov, G.V. (Ed.) (1968). *Handbook of the physiochemical properties of the elements*. Oldbourne.
An extensive compilation of data from Western and Eastern sources. Contains a number of inaccuracies, but also contains a large quantity of data on the rarer elements, hard to find elsewhere.

Gschneidner, K.A. (1964). Physical properties and interrelationships of metallic and semimetallic elements. *Solid State Physics*, *16*, 275–426.
Probably the best source of its time, this work is very well referenced, and full explanations are given of estimated or approximate data.

Internet sources

Winter, M. WebElements. University of Sheffield, *www.webelements.com*
A comprehensive source of information on all the elements in the Periodic Table. If it has a weakness, it is in the definitions and values of some mechanical properties.

Martindale's: physics web pages, *www.martindalecenter.com/Calculators3A.html*

Aluminum alloys

Aluminum standards and data. (1990). The Aluminum Association.

The properties of aluminum and its alloys. (1981). The Aluminum Federation.

Technical data sheets. (1993). ALCAN International Ltd.

Technical data sheets. (1993). ALCOA.

Technical data sheets. (1994). Aluminum Pechiney, France.

Internet sources

Aluminum Federation, *www.alfed.org.uk*

Aluminum World, *www.sovereign-publications.com/aluminum.htm*

International Aluminum Institute, *www.world-aluminum.org*

Babbitt Metal

The term "Babbitt Metal" denotes a series of lead-tin-antimony–bearing alloys, the first of which was patented in the United States by Isaac Babbitt in 1839. Subsequent alloys are all variations on his original composition.

ASTM Standard B23-83: White metal bearing alloys (known commercially as "Babbitt Metal"), *ASTM Annual Book of Standards*, Vol. 02.04.

Beryllium

Designing with beryllium (1996). Brush Wellman.

Beryllium optical materials (1996). Brush Wellman.

Internet sources

Brush Wellman, *www.brushwellman.com*

Cadmium

Cadmium production, properties and uses (1991). International Cadmium Association.

Chromium

Castings, chromium-nickel alloy, ASTM Standard A560-89: *ASTM Annual Book of Standards, 01*(02).

Cobalt alloys

Betteridge, W. (1982). *Cobalt and its alloys*. Ellis Horwood.
A good general introduction to the subject.

Columbium alloys

See Niobium alloys.

Copper alloys

ASM Metals Handbook (10th ed.) (1990). ASM International

West, E.G. (1979). *The selection and use of copper-rich alloys*. Oxford University Press for the Design Council, the British Standards Institution, and the Council of Engineering Institutions.

Copper Development Association data sheets, 26 (1988), 27 (1981), 31 (1982), 40 (1979), and 82 (1982). Copper Development Association, and The Copper Development Association.

Megabytes on coppers (1994). The Copper Development Association and Granta Design Limited.

Smithells metals reference book, (7th ed.). (1992). E.A. Brandes & G.B. Brook (Eds.). Butterworth-Heinemann.

Internet sources

Copper Development Association, *www.cda.org.uk*

Copper page, *www.copper.org*

Gold and dental alloys

Gold: Art, science and technology (1992). *Interdisciplinary Science Reviews*, *17*(3). The Institute of Materials, ISSN 0308-0188.

Focus on gold (1992). *Interdisciplinary Science Reviews*, *17*(4). The Institute of Materials, ISSN 0308-0188.

ISO Standard 1562: Dental casting gold alloys (1993). International Standards Organization.

ISO Standard 8891: Dental casting alloys with Noble metal content of 25% up to but not including 75% (1993). International Standards Organization.

Internet sources

Goodfellow Metals, *www.goodfellow.com*
Supplier of pure and precious metals, mainly for laboratory use. Website contains price and property information for its entire stock.

Jeneric Pentron, Casting alloys, *www.jeneric.com/casting*
An informative commercial site. An extensive source of information, both for natural biological materials and for metals used in dental treatments.

Rand Refinery Limited, Chamber of Mines website, *www.bullion.org.za/associates/rr.htm*
Contains useful information on how gold is processed to varying degrees of purity.

Indium

The Indium Info Center, Indium Corp. of America. *www.indium.com*

Lead

ASTM Standard B29-79: Pig lead. *ASTM Annual Book of Standards*, *02*(04).

ASTM Standard B102-76: Lead- and tin-alloy die castings. *ASTM Annual Book of Standards*, *02*(04).

ASTM Standard B749-85: Lead and lead alloy strip, sheet, and plate products, *ASTM Annual Book of Standards*, *02*(04).

Lead for corrosion resistant applications. Lead Industries Association.

ASM metals handbook (9th ed.). (1986). Vol. 2, 500–510.
See also Babbitt Metal (listed previously).

Internet sources

Lead Development Association International, *www.ldaint.org*

India Lead Zinc Development Association, *www.ilzda.com*

Magnesium alloys

Technical data sheets (1994). Magnesium Elektron Ltd.

Technical literature (1994). Magnesium Corp. of America.

Molybdenum

ASTM Standard B386-85: Molybdenum and molybdenum alloy plate, sheet, strip and foil. *ASTM Annual Book of Standards*, *02*(04).

ASTM Standard B387-85: Molybdenum and molybdenum alloy bar, rod and wire. *ASTM Annual Book of Standards*, *02*(04).

Nickel

A major data source for nickel and its alloys is the Nickel Development Institute (NIDI), a global organization with offices on every continent except Africa. NIDI freely gives away large quantities of technical reports and data compilations, not only for nickel and high-nickel alloys but also for other nickel-bearing alloys, such as stainless steel.

ASTM Standard A297-84: Steel castings, iron-chromium and iron-chromium-nickel, heat resistant, for general application. *ASTM Annual Book of Standards*, *01*(02).

ASTM Standard A344-83, Drawn or rolled nickel-chromium and nickel-chromium-iron alloys for electrical heating elements. *ASTM Annual Book of Standards*, *02*(04).

ASTM Standard A494-90: Castings, nickel and nickel alloy. *ASTM Annual Book of Standards*, *02*(04).

ASTM Standard A753-85: Nickel-iron soft magnetic alloys. *ASTM Annual Book of Standards*, *03*(04).

Betteridge, W. (1984). *Nickel and its alloys*. Ellis Horwood.
A good introduction to the subject.

INCO (1995). *High-temperature, high-strength nickel base alloys*. Nickel Development Institute.
Tabular data for over 80 alloys.

Elliott, P. (1990). *Practical guide to high-temperature alloys*. Nickel Development Institute.

INCO (1978). *Heat & corrosion resistant castings*. Nickel Development Institute.

INCO (1969). *Engineering properties of some nickel copper casting alloys*. Nickel Development Institute.

INCO (1968). *Engineering properties of IN-100 alloy*. Nickel Development Institute. INCO (1969). *Engineering properties of nickel-chromium alloy 610 and related casting alloys*. Nickel Development Institute.

INCO (1968). *Alloy 713C: technical data*. Nickel Development Institute.

INCO (1981). *Alloy IN-738: technical data*. Nickel Development Institute.

INCO (1976). *36% nickel-iron alloy for low temperature service.* Nickel Development Institute.

ASTM Standard A658 (discontinued 1989): Pressure vessel plates, alloy steel, 36 percent nickel. *ASTM Annual Book of Standards*, pre-1989 editions.

ASM Metals Handbook (9th ed.) (1986). Vol. 3, 125–178.

Internet sources

Carpenter Technology Corp, *www.cartech.com*

Nickel Development Institute, *www.nidi.org*

Special Metals Corp., *www.specialmetals.com*

Steel & nickel based alloys, *www.superalloys.co.uk*

Niobium (Columbium) alloys

ASTM Standard B391-89: Niobium and niobium alloy ingots. *ASTM Annual Book of Standards*, *02*(04).

ASTM Standard B392-89: Niobium and niobium alloy bar, rod and wire. *ASTM Annual Book of Standards*, *02*(04).

ASTM Standard B393-89: Niobium and niobium alloy strip, sheet and plate. *ASTM Annual Book of Standards*, *02*(04).

ASTM Standard B652-85: Niobium-hafnium alloy ingots. *ASTM Annual Book of Standards*, *02*(04).

ASTM Standard B654-79: Niobium-hafnium alloy foil, sheet, strip and plate. *ASTM Annual Book of Standards*, *02*(04).

ASTM Standard B655-85: Niobium-hafnium alloy bar, rod and wire. *ASTM Annual Book of Standards*, *02*(04).

Internet sources

Husted, R., Los Alamos National Laboratory. *www-c8.lanl.gov/infosys*

An overview of Niobium and its uses.

Palladium

ASTM Standard B540-86: Palladium electrical contact alloy. *ASTM Annual Book of Standards*, *03*(04).

ASTM Standard B563-89: Palladium-silver-copper electrical contact alloy. *ASTM Annual Book of Standards*, *03*(04).

ASTM Standard B589-82: Refined palladium. *ASTM Annual Book of Standards*, *02*(04).

ASTM Standard B683-90: Pure palladium electrical contact material. *ASTM Annual Book of Standards*, *03*(04).

ASTM Standard B685-90: Palladium-copper electrical contact material. *ASTM Annual Book of Standards*, *03*(04).

ASTM Standard B731-84: 60% palladium-40% silver electrical contact material. *ASTM Annual Book of Standards*, *03*(04).

Internet sources

Jeneric Pentron, Casting alloys. *www.jeneric.com/casting*
An informative commercial site, limited to dental alloys.

Goodfellow Metals, *www.goodfellow.com*
Supplier of pure and precious metals, mainly for laboratory use. Website contains price and property information for its entire stock.

Platinum alloys

ASTM Standard B684-81: Platinum-iridium electrical contact material. *ASTM Annual Book of Standards*, *03*(04).

Elkonium Series 400 datasheets, CMW.

ASM Metals Handbook (9th ed.) (1986). Vol. 2, 688–698.

Silver alloys

ASTM Standard B413-89: Refined silver. *ASTM Annual Book of Standards*, *02*(04).

ASTM Standard B617-83: Coin silver electrical contact alloy. *ASTM Annual Book of Standards*, *03*(04).

ASTM Standard B628-83: Silver-copper eutectic electrical contact alloy. *ASTM Annual Book of Standards*, *03*(04).

ASTM Standard B693-87: Silver-nickel electrical contact materials. *ASTM Annual Book of Standards*, *03*(04).

ASTM Standard B742-90: Fine silver electrical contact fabricated material. *ASTM Annual Book of Standards*, *03*(04).

ASTM Standard B780-87: 75% silver, 24.5% copper, 0.5% nickel electrical contact alloy. *ASTM Annual Book of Standards*, *03*(04).

Elkonium Series 300 datasheets (1996). CMW.

Elkonium Series 400 datasheets (1996). CMW.

Internet sources

The Silver Institute, *www.silverinstitute.org*

Jeneric Pentron, Casting alloys. *www.jeneric.com/casting*
An informative commercial site, limited to dental alloys.

Goodfellow Metals, *www.goodfellow.com*
Supplier of pure and precious metals, mainly for laboratory use. Website contains price and property information for its entire stock.

Tantalum alloys

ASTM Standard B365-86: Tantalum and tantalum alloy rod and wire. *ASTM Annual Book of Standards*, *02*(04).

ASTM Standard B521-86: Tantalum and tantalum alloy seamless and welded tubes. *ASTM Annual Book of Standards*, *02*(04).

ASTM Standard B560-86: Unalloyed tantalum for surgical implant applications. *ASTM Annual Book of Standards*, *15*(01).

ASTM Standard B708-86: Tantalum and tantalum alloy plate, sheet and strip. *ASTM Annual Book of Standards*, *02*(04).

Tantalum data sheet (1996). The Rembar Company.

ASM Handbook (9th ed.) (1986). Vol. 3, 323–325, 343–347.

Tin alloys

ASTM Standard B32-89: Solder metal. *ASTM Annual Book of Standards*, *02*(04).

ASTM Standard B339-90: Pig tin. *ASTM Annual Book of Standards*, *02*(04).

ASTM Standard B560-79: Modern pewter alloys. *ASTM Annual Book of Standards*, *02*(04).

Barry, B.T.K. & Thwaites, C.J. (1983). *Tin and its alloys and compounds*. Ellis Horwood.

ASM Metals Handbook (9th ed.) (1986). Vol. 2, 613–625
See also Babbitt Metal (earlier).

Tin Research Association, *www.tintechnology.com*

Titanium alloys

Technical data sheets (1993). Titanium Development Association.

Technical data sheets (1993). The Titanium Information Group.

Technical data sheets (1995). IMI Titanium.

Internet sources

Titanium Information Group, *www.titaniuminfogroup.co.uk*

The International Titanium Association, *www.titanium.org*
This association site has a large list of member companies and comprehensive information on titanium and its alloys.

Tungsten alloys

ASTM Standard B777-87: Tungsten base, high-density metal. *ASTM Annual Book of Standards*, *02*(04).

Yih, S.W.H. & Wang, C.T. (1979). *Tungsten*. Plenum Press.

ASM Metals Handbook (9th ed.) (1986). Vol. 7, p. 476.

Tungsten data sheet (1996). The Rembar Company.

Royal Ordnance speciality metals datasheet (1996). British Aerospace Defence Ltd.

CMW datasheets (1996). CMW.

Internet sources

North American Tungsten, *www.northamericantungsten.com*

Uranium

Uranium Information Centre, Australia, *www.uic.com.au*

UXC Jan 96 uranium indicator update, *www.uxc.com/review/uxc_prices/aspx*

Vanadium

Teledyne Wah Chang (1996). *Vanadium brochure*. TWC .

Zinc

ASTM Standard B6-87: Zinc. *ASTM Annual Book of Standards*, *02*(04).

ASTM Standard B69-87: Rolled zinc. *ASTM Annual Book of Standards*, *02*(04).

ASTM Standard B86-88: Zinc-alloy die castings. *ASTM Annual Book of Standards*, *02*(02).

ASTM Standard B418-88: Cast and wrought galvanic zinc anodes. *ASTM Annual Book of Standards*, *02*(04).

ASTM Standard B791-88: Zinc-aluminum alloy foundry and die castings. *ASTM Annual Book of Standards*, *02*(04).

ASTM Standard B792-88: Zinc alloys in ingot form for slush casting. *ASTM Annual Book of Standards*, *02*(04).

ASTM Standard B793-88: Zinc casting alloy ingot for sheet metal forming dies. *ASTM Annual Book of Standards*, *02*(04).

Goodwin, F.E., & Ponikvar, A.L. (Eds.) (1989). *Engineering properties of zinc alloys* (3rd ed.). International Lead Zinc Research Organization.
An excellent compilation of data, covering all industrially important zinc alloys.

Chivers, A.R.L. (1981). Zinc diecasting. *Engineering Design Guide*, 41, OUP.
A good introduction to the subject.

ASM Metals Handbook (9th ed.) (1986). Properties of zinc and zinc alloys. Vol.2, 638–645.

Internet sources

International Zinc Association, *www.iza.com*

Zinc Industrias Nacionales S.A.-Peru, *www.zinsa.com*

Zinc: Eastern Alloys, *www.eazall.com*

India Lead Zinc Development Association, *www.ilzda.com*

Zirconium

ASTM Standard B350-80: Zirconium and zirconium alloy ingots for nuclear application. *ASTM Annual Book of Standards, 02*(04).

ASTM Standards B352-85, B551-83, and B752-85: Zirconium and zirconium alloys. *ASTM Annual Book of Standards, 02*(04).

Teledyne Wah Chang (1996). *Zircadyne: properties & applications*. TWC

ASM Metals Handbook (9th ed.) (1986). Vol. 2, 826–831.

D2.4 Ferrous metals

Ferrous metals are probably the most thoroughly researched and documented class of materials. Nearly every developed country has its own system of standards for irons and steels. Recently, continental and worldwide standards have been developed, which have achieved varying levels of acceptance. There is a large and sometimes confusing literature on the subject. This section is intended to provide the user with a guide to some of the better information sources.

Ferrous metals, general data sources

Bringas, J.E. (Ed.) (1995). *The metals black book – ferrous metals* (2nd ed.). CASTI Publishing.
An excellent short reference work.

ASM Metals Handbook (10th ed.) (1990). Vol. 1. ASM International.
Authoritative reference work for North American irons and steels.

ASM Metals Handbook (desk ed.) (1985). ASM International.
A summary of the multi-volume ASM Metals Handbook.

Wegst, C.W. (2010) *Stahlschlüssel* (in English: *Key to steel*). Verlag Stahlschlüssel Wegst GmbH.
Published every three years, in German, French, and English. Excellent coverage of European products and manufacturers.

Woolman, J., & Mottram, R.A. (1966). *The mechanical and physical properties of the British standard in steels*. Pergamon Press.
Still highly regarded, but based around a British Standard classification system that has been officially abandoned.

Brandes, E.A., & Brook, G.R. (Eds.) (1992). *Smithells metals reference book* (7th ed.). Butterworth-Heinemann.
An authoritative reference work, covering all metals.

Waterman, N.A. & Ashby, M.F. (Eds.) (1996). *Materials Selector*. Chapman and Hall.
Covers all materials; irons and steels are in vol. 2.

Sharpe, C. (Ed.) (1993). *Kempe's engineering year book* (98th ed.). Benn.
Updated each year; has good sections on irons and steels.

Irons and steels standards

Increasingly, national and international standards organizations are providing a complete catalog of their publications on the web. Two of the most comprehensive printed sources are listed here.

Iron and steel specifications (9th ed.) (1998). British Iron and Steel Producers Association (BISPA).
Comprehensive tabulations of data from British Standards on irons and steels, as well as some information on European and North American standards. The same information is available on a searchable CD.

ASTM Annual Book of Standards, Vols. 01(01) through 01(07).
The most complete set of American irons and steels standards. Summaries of the standards can be found on the web at www.astm.org

Cross-referencing of similar international standards and grades

It is difficult to match, even approximately, equivalent grades of iron and steel between countries. No coverage of this subject can ever be complete, but the references listed here are helpful.

Gensure, J.G., & Potts, D.L. (1988). *International metallic materials cross reference* (3rd ed.). Genium Publishing.
Comprehensive worldwide coverage of the subject, well indexed.

Bringas, J.E. (Ed.) (1995). *The metals black book–ferrous metals* (2nd ed.). CASTI Publishing.
Easy-to-use tables for international cross-referencing. (*See "General" section for more information.*)

Unified numbering system for metals and alloys (2nd ed.), (1977). Society of Automotive Engineers.
An authoritative reference work, providing a unifying structure for all standards published by U.S. organizations. No coverage of the rest of the world.

Iron and steel specifications (7th ed.) (1989). British Steel.
Lists "Related Specifications" for France, Germany, Japan, Sweden, U.K. and U.S.A.

Cast irons

Scholes, J.P. (1979). *The selection and use of cast irons*. Engineering Design Guides, OUP.

Angus, H.T. (1976). *Cast iron: physical and engineering properties*. Butterworths.

Cast irons, American standards

These can all be found in the *Annual Book of ASTM Standards*, *01*(02).

ASTM A220M-88: Pearlitic malleable iron, *www.astm.org*

ASTM A436-84: Austenitic gray iron castings; *www.astm.org*

ASTM A532: Abrasion-resistant cast irons; *www.astm.org*

ASTM A602-70 (reapproved 1987): Automotive malleable iron castings.

Cast irons, international standards

These are available from the ISO Central Secretariat, Geneva, Switzerland.

ISO 185:1988 Grey cast iron – classification.

ISO 2892:1973 Austenitic cast iron.

ISO 5922:1981 Malleable cast iron.

Cast irons, British standards

Compared with steels, there are relatively few standards for cast iron, which makes it feasible to list them all. Standards are available from BSI Customer Services, London.

BS 1452:1990 Flake graphite cast iron.

BS 1591:1975 Specification for corrosion resisting high silicon castings.

BS 2789:1985 Iron castings with spheroidal or nodular graphite.

BS 3468:1986 Austenitic cast iron.

BS 4844:1986 Abrasion resisting white cast iron.

BS 6681:1986 Specification for malleable cast iron.

Carbon and low alloy steels

ASM metals handbook (10th ed.) (1990). Vol. 1. ASM International.
Authoritative reference work for North American irons and steels.

Fox, J.H.E. (1980). An introduction to steel selection: part 1, carbon and low-alloy steels. *Engineering Design Guide*, no. 34. Oxford University Press.

Stainless steels

ASM metals handbook (10th ed.) (1990). Vol. 1. ASM International.
Authoritative reference work for North American irons and steels.

Elliott, D., & Tupholme, S.M. (1981). An introduction to steel selection: part 2, stainless steels. *Engineering Design Guide*, no. 43. Oxford University Press.

Peckner, D., & Bernstein, I.M. (1977). *Handbook of stainless steels*. McGraw-Hill.

Design guidelines for the selection and use of stainless steel (1991). *Designers' Handbook Series*, no. 9014. Nickel Development Institute.
The Nickel Development Institute (NIDI) is a worldwide organization that gives away a large variety of free literature about nickel-based alloys, including stainless steels. NIDI European Technical Information Centre.

General Internet sites for ferrous metals

British Constructional Steel Work Association, *www.steelconstruction.org*

International Iron & Steel Institute, *www.worldsteel.org*

Iron & Steel Trades Confederation, *www.istc-tu.org*

National Association of Steel Stock Holders, *www.nass.org.uk*

Steel Manufacturers Association, *www.steelnet.org*

Steel: Bethlehem Steel, *www.bethsteel.com* (steel)

Steel: Corus Group, *www.corusgroup.com(steel)*

Steel: Automotive Steel Library, *www.autosteel.org(steel)*

Steel: Great Plains Stainless, *www.gpss.com(steel)*

SteelSpec, *www.steelspec.org.uk/index.htm(steel)*

D2.5 Polymers and elastomers

Polymers are not subject to the same strict specifications as metals. Data tend to be producer-specific. Sources are consequently scattered, incomplete, and poorly presented. Saechtling is the best; although no single hard-copy source is completely adequate, all those listed here are worth consulting. See also Section D.4, Databases and Expert Systems as software; some are good on polymers.

Saechtling, H. (Ed.) (1983). *Saechtling: International plastics handbook*. MacMillan.
The most comprehensive of the hard-copy data sources for polymers.

Seymour, R.B. (1987). *Polymers for engineering applications*. ASM International.
Property data for common polymers. A starting point, but insufficient detail for accurate design or process selection.

Murphy, J (Ed.) (1991). *New horizons in plastics, a handbook for design engineers*. WEKA Publishing.

ASM engineered materials handbook, Vol 2. engineering plastics (1989). ASM International.

Harper, C.A. (Ed.) (1975). *Handbook of plastics and elastomers*. McGraw-Hill.

International plastics selector, plastics (9th ed.) (1987). International Plastics Selector.

Dominighaus, H. (Ed.) (1992). *Die kunststoffe and ihre eigenschaften*. VDI Verlag.

van Krevelen, D.W. (1990). *Properties of polymers* (3rd ed.). Elsevier.
Correlation of properties with structure; estimation from molecular architecture.

Bhowmick, A.K., & Stephens, H.L. (1988). *Handbook of elastomers*. Marcel Dekker.

ICI technical service notes (1981). ICI Plastics Division, Engineering Plastics Group.

Technical data sheets (1995). Malaysian Rubber Producers Research Association.
Data sheets for numerous blends of natural rubber.

Ossvald, T.A., Baur, E., Brinkmann, F., Oberbach, K., & Schmachtenberg, E. (2006). *International plastics handbook*, Carl Hanser Verlag.

Software and Internet sources

CAMPUS Plastics database, *www.campusplastics.com*

CES Polymers database, *www.grantadesign.com*

GE Plastics, *www.ge.com/en/company/businesses/ge_plastics.htm*

Harboro Rubber Co., *www.harboro.co.uk*

IDES Resin Source, *www.ides.com*

MERL, *www.merl-ltd.co.uk*

Plastics.com, *www.plastics.com*

Matweb.com, *www.matweb.com*

M-base engineering, *www.m-base.de/main*

D2.6 Ceramics and glasses

Sources of data for ceramics and glasses, other than the suppliers' data sheets, are limited. Texts and handbooks—for example, the ASM's (1991) *Engineered Materials Handbook, Vol. 4*, Morell's (1985) compilations, Neville's (1996) book on concrete, Boyd and Thompson's (1980) *Handbook on Glass*, and Sorace's (1996) treatise on stone are useful starting points. The CES Ceramics Database contains recent data for ceramics and glasses. In the end, however, it is the manufacturer to whom one has to turn: The data sheets for their products are the most reliable sources of information.

Ceramics and ceramic-matrix composites

ASM engineered materials handbook, Vol. 4, Ceramics and glasses (1991). ASM International.

Waterman, N., & Ashby, M.F. (Eds.) (1996). *Materials selector*. Chapman and Hall.

Brook, R.J. (Ed.) (1991). *Concise encyclopedia of advanced ceramic materials*. Pergamon Press.

Cheremisinoff, N.P. (Ed.) (1990). *Handbook of ceramics and composites*, 3 Vols. Marcel Dekker.

Clark, S.P. (Ed.) (1996). *Handbook of physical constants*, memoir 97. Geological Society of America.

Schwartz, M.M. (1992). *Handbook of structural ceramics*. McGraw-Hill.
A great deal of data and information on processing and applications.

Kaye, G.W.C., & Laby, T.H. (1986). *Tables of physical & chemical constants* (15th ed.). Longman.

Kingery, W.D., Bowen, H.K., & Uhlmann, D.R. (1976). *Introduction to ceramics* (2nd ed.). Wiley.

Materials Engineering (1992). *Materials selector*. Penton Press.

Morrell, R. (1985). *Handbook of properties of technical & engineering ceramics, Parts I and II*. National Physical Laboratory.

Musikant, S. (1991). *What every engineer should know about ceramics*. Marcel Dekker.
Good on data.

Richerson, D.W. (1992). *Modern ceramic engineering* (2nd ed.). Marcel Dekker.

Brandes, E.A., & Brook, G.B. (Eds.) (1992). *Smithells metals reference book* (7th ed). Butterworth-Heinemann.

Harper, C.A. (Ed.) (2001). *Handbook of ceramics, glasses and diamonds*. McGraw-Hill.
A comprehensive compilation of data and design guidelines.

Richerson, D.W. (2000). *The magic of ceramics*. American Ceramics Society.
A readable introduction to ceramics, both old and new.

Glasses

ASM engineered materials handbook, Vol. 4, Ceramics and glasses (1991). ASM International.

Boyd D.C., & Thompson D.A. (1980). *Glass*. Reprinted from Kirk-Othmer: *Encyclopedia of chemical technology, Vol. 11* (3rd ed.), pp 807–880. Wiley.

Oliver, D.S. (1975). *Engineering design guide 05: The use of glass in engineering*. Oxford University Press.

Bansal, N.P., & Doremus, R.H. (1966). *Handbook of glass properties*. Academic Press.

Cement and concrete

Cowan, H.J., & Smith, P.R. (1988). *The science and technology of building materials*. Van Nostrand-Reinhold.

Illston, J.M., Dinwoodie, J.M., & Smith, A.A. (1979). *Concrete, timber and metals*. Van Nostrand-Reinhold.

Neville, A.M. (1996). *Properties of concrete* (4th ed.). Longman Scientific and Technical.
An excellent introduction to the subject.

D2.7 Composites: PMCs, MMCs, and CMCs

The fabrication of composites allows so many variants that no hard-copy data source can capture them all; instead, they list properties of matrix and reinforcement and of certain generic lay-ups or types. The *Engineers Guide* and the *Composite Materials Handbook*, listed first, are particularly recommended.

Composites, general

Weeton, J.W., Peters, D.M., & Thomas, K.L. (Eds.) (1987). *Engineers guide to composite materials*. ASM International.
The best starting point: data for all classes of composites.

Schwartz, M.M. (Ed.) (1992). *Composite materials handbook* (2nd ed.). McGraw-Hill.
Lots of data on PMCs, less on MMCs and CMCs, processing, fabrication, applications, and design information.

ASM engineered materials handbook, Vol 1: Composites (1987). ASM International.

Seymour, R.B. (1991). *Reinforced plastics, properties and applications*. ASM International.

Cheremisinoff, N.P. (Ed.) (1990). *Handbook of ceramics and composites, Vols. 1–3*. Marcel Dekker.

Kelly, A. (Ed.) (1989). *Concise encyclopedia of composited materials*. Pergamon Press.

Middleton, D.H. (1990). *Composite materials in aircraft structures*. Longman Scientific and Technical Publications.

Smith, C.S. (1990). *Design of marine structures in composite materials*. Elsevier Applied Science.

Metal matrix composites

See, first, the sources listed under "All Composite Types; for more detail, go to

ASM engineered materials handbook, Vol. 1: Composites (1987). ASM International.

Technical data sheets (1995). Duralcan USA.

Technical data sheets (1995). 3M Company.

D2.8 Foams and cellular solids

Many of the references given in Section D2.5 for polymers and elastomers mention foam. The references given here contain much graphical data and simple formulae that allow properties of foam to be estimated from the foam's density and the properties of the solid of which it is made, but in

the end it is necessary to contact suppliers. See also Section D.4, Databases and Expert Systems as Software; some are good on foams. For woods and wood-based composites, see Section D2.10.

Cellular polymers (1981–1996). RAPRA Technology.

Encyclopedia of chemical technology, Vol. 2 (3rd ed.), pp. 82–126 (1980). Wiley.

Encyclopedia of polymer science and engineering, Vol. 3 (2nd ed.), section C (1985). Wiley.

Gibson, L.J., & Ashby, M.F. (1997). *Cellular solids.* Cambridge University Press.
Basic text on foamed polymers, metals, ceramics, and glasses, as well as and natural cellular solids.

Handbook of industrial materials (2nd ed.), pp. 537–556 (1992). Elsevier Advanced Technology.

Hilyard, N.C., & Cunningham, A. (Eds.) (1994). *Low density cellular plastics —physical basis of behaviour*. Chapman and Hall.
Specialized articles on aspects of polymer-foam production, properties, and uses.

Plascams, Version 6 (1995). *Plastics computer-aided materials selector*. RAPRA Technology Limited.

Saechtling, H. (Ed.) (1983). *Saechtling: International plastics handbook*. MacMillan.

Seymour, R.P. (1987). *Polymers for engineering applications*. ASM International.

D2.9 Stone, rocks, and minerals

There is an enormous literature on rocks and minerals. Start with the handbooks listed below; then ask a geologist for guidance.

Atkinson B.K. (1987). *The fracture mechanics of rock*. Academic Press.

Clark, S.P., Jr. (Ed.) (1966). *Handbook of physical constants*. Memoir 97, The Geological Society of America.
Old but trusted compilation of property data for rocks and minerals.

Lama, R.E., & Vutukuri, V.S. (Eds.) (1978). *Handbook on mechanical properties of rocks, Vols. 1–4*. Trans Tech Publications.

Griggs, D., & Handin, J. (Eds.) (1960). *Rock deformation*. Memoir 79, The Geological Society of America.

Sorace, S. (1966). Long-term tensile and bending strength of natural building stones. *Materials and Structures, 29*, 426–435.

D2.10 Timber woods and wood-based composites

Woods, like composites, are anisotropic; useful sources list properties along and perpendicular to the grain. The U.S. Forest Products Laboratory's *Wood Handbook* and Kollmann and Coté's *Principles of Wood Science and Technology* are particularly recommended.

Woods and timbers, general

Bodig, J., & Jayne, B.A. (1982). *Mechanics of wood and wood composites*. Van Nostrand Reinholt.

Dinwoodie, J.M. (1989). *Wood, nature's cellular polymeric fiber composite*. The Institute of Metals.

Dinwoodie, J.M. (1981). *Timber, its nature and behaviour*. Van Nostrand-Reinhold.
Basic text on wood structure and properties. Not much data.

Gibson, L.J., & Ashby, M.F. (1997). *Cellular solids* (2nd ed.). Cambridge University Press.

Jane, F.W. (1970). *The structure of wood* (2nd ed.). A. and C. Black.

Kollmann, F.F.P., & Côté, W.A., Jr. (1968). *Principles of wood science and technology, Vol. 1, Solid wood*. Springer-Verlag.
The bible.

Kollmann, F., Kuenzi, E., & Stamm, A. (1968). *Principles of wood science and technology, Vol. 2, Wood-based materials*. Springer-Verlag.

Schniewind, A.P. (Ed.) (1989). *Concise encyclopedia of wood and wood-based materials*. Pergamon Press.

Timber: Data compilations

BRE (1996). *BRE information papers*. Building Research Establishment (BRE).

Forest Products Laboratory (1989). *Handbook of wood and wood-based materials*. Forest Service, U.S. Department of Agriculture. Hemisphere Publishing Corporation.
A massive compilation of data for North American woods.

Informationsdienst Holz (1996). Merkblattreihe Holzarten.

TRADA (1978/1979). *Timbers of the world, Vols. 1– 9*. Timber Research and Development Association.

TRADA (1991). *Information sheets*. Timber Research and Development Association.

Timber and wood-composite standards

Great Britain

British Standards Institution (BSI), *www.bsigroup.com*

Germany

Deutsches Institut für Normung (DIN), *www.din.de*

United States

American Society for Testing and Materials (ASTM), *www.astm.org*

Software and Internet data sources

CES woods database, *www.grantadesign.com*

PROSPECT, Oxford Forestry Institute, Department of Plant Sciences, Oxford University *www.plants.ox.ac.uk/ofi/prospect/index*
A database of the properties of tropical woods of interest to a wood user; includes information about uses, workability, treatments, and origins.

Woods of the world (1994). Tree Talk.
A CD-ROM of woods, with illustrations of structure, information about uses, origins, habitat, and so on.

WoodWeb, *www.woodweb.com*
Woodwork industry web service. Contains information about woodworking-related companies.

D2.11 Natural fibers and other natural materials

Natural materials, general

Alexander, R.M. (1983). *Animal mechanics* (2nd ed.). Blackwell Scientific.

Brown, C.H. (1975). *Structural materials in animals*. Pitman.

Currey, J.D., Wainwright, S.A., & Biggs, W.D. (1982). *Mechanical design in organisms*. Princeton University Press.

Fung, Y.C. (1993). *Biomechanics: Mechanical properties of living tissues* (2nd ed.). Springer-Verlag.

Gibson, L.J., & Ashby, M.F. (1997). *Cellular solids: structure and properties* (2nd ed.). Cambridge University Press.

McMahon, T.A. (1984). *Muscles, reflexes, and locomotion*. Princeton University Press.

Silver, F.H. (1987). *Biological materials: Structure, mechanical properties, and modeling of soft tissues*. New York University Press.

Vincent, J.F.V., & Currey, J.D. (1980). The mechanical properties of biological materials. *Proceedings of the Symposia of the Society for Experimental Biology* (no.34). Cambridge University Press for the Society for Experimental Biology.

Vogel, S., & Calvert, R.A (illustrator) (1988). *Life's devices: The physical world of animals and plants*. Princeton University Press.

Yamada, H. (1970). *Strength of biological materials*. Williams & Wilkins.

Wood and wood-like materials (wood cell wall, wood, cork, bamboo, palm, rattan)

Amada, S., Munekata, T., Nagase, Y., Ichikawa, Y., Kirigai, A., & Yang, Z.F. (1996). The mechanical structures of bamboos in viewpoint of functionally gradient and composite materials, *J. Compos. Mater.*, *30*(7), 800–819.

Bhat, K., & Mathew, A. (1995). Structural basis of rattan biomechanics. *Biomimetics*, *3*(2), 67–80.

Frühwald, A., Peek, R.-D., & Schulte, M. (1992). Nutzung von kokospalmenholz am beispiel von nordsulawesi, Indonesien, technical report. *Mitteilungen der Bundesforschungsanstalt für Forst- und Holzwirtschaft, 171*. Kommissionsverlag Max Wiederbusch.

Gibson, L.J., Easterling, K.E., & Ashby, M.F. (1981). The structure and mechanics of cork. *P. Roy. Soc. Lond. A*, *377*(1769), 99–117.

Godbole, V.S., & Lakkad, S.C. (1986). Effect of water-absorption on the mechanical properties of bamboo. *J. Mater. Sci. Lett.*, *5*(3), 303–304.

Janssen, J.J.A. (1991). Mechanical properties of bamboo *Forestry Sciences*, Vol. 37, Kluwer Academic Publishers.

Killmann, W. (1983). Some physical-properties of the coconut palm stem. *Wood Sci. Tech.*, *17*(3), 167–185.

Killmann, W. (1993). *Struktur, eigenschaften and nutzung von stämmen wirtschaftlich wichtiger palmen*. PhD thesis, Universität Hamburg.

Kloot, N. (1952). Mechanical and physical properties of coconut palm. *Austr. J. Appl. Sci.*, *3*(4), 293–322.

Lakkad, S.C., & Patel, J.M. (1981). Mechanical properties of bamboo, a natural composite. *Fibre Sci. Tech.*, *14*(4), 319–322.

Mark, R.E. (1967). *Cell wall mechanics of tracheids*. Yale University Press.

Rich, P.M. (1987). Mechanical structure of the stem of arborescent palms. *Bot. Gazette*, *148*(1), 42–50.

Rosa, M.E., & Fortes, M.A. (1988a). Stress-relaxation and creep of cork. *J. Mater. Sci.*, *23*(1), 35–42.

Rosa, M.E., & Fortes, M.A. (1988b). Rate effects on the compression and recovery of dimensions of cork. *J. Mater. Sci.*, *23*(3), 879–885.

Rosa, M.E., & Fortes, M.A. (1991). Deformation and fracture of cork in tension. *J. Mater. Sci.*, *26*(2), 341–348.